WAZUH SOC ANALYST HOME LAB — TECHNICAL REPORT

ROGUE DOMAIN ADMIN DETECTION AND INVESTIGATION WITH WAZUH

Persistence and privilege escalation in a Windows Server 2019 Active Directory home lab: investigation, evidence, custom detection rule and a beginner-reproducible procedure

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Lab domain	HopeLabs.Local (NetBIOS: HOPELABS)
Tools	Wazuh (dashboard/manager, 4.x index), Windows Server 2019 AD, Windows 10, Sysmon, PowerShell, Splunk 9.4.2 (installed, not used for findings)
Classification	Training lab — all accounts and credentials are disposable lab values

NOTE. Every claim in this report is tied to a numbered figure (evidence ID E-nn), a command card (C-n) or a query card (Q-n). Where something was **not** verified, the report says so. The original screenshots are untouched; the figures are presentation copies (see Chapter 9).

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Abbreviations and Acronyms

Table 1. Abbreviations used in this report

Term	Meaning
AD / AD DS	Active Directory (Domain Services): Microsoft's central directory of users, computers and groups
DC	Domain Controller: the server that holds Active Directory and checks logins
DNS	Domain Name System: translates names (HopeLabs.Local) into IP addresses
DQL	Dashboards Query Language: the simple search syntax in the Wazuh dashboard
SIEM	Security Information and Event Management: collects and analyses security logs
SID	Security Identifier: the unique number Windows uses for an account or group
SOC	Security Operations Center
UAC	User Account Control (also the account-control flag value in event 4738)
MITRE ATT&CK	Public catalogue of attacker techniques (T-numbers)
Sysmon	Microsoft System Monitor: logs process creation and other endpoint activity
Evidence ID (E-nn)	Identifier of a figure in this report; the register is in Chapter 9

Chapter 1 — Executive Summary

Purpose. The Account Compromise Investigation lab of the Wazuh SOC Analyst program asks an analyst to investigate account creation and privilege changes (Figure 1). In this lab I created a disposable account on my Domain Controller, promoted it to **Domain Admins**, signed in with it on a workstation, and then investigated the activity in Wazuh as a SOC analyst who does not know what happened.

Security problem. Attackers who gain admin access often create a new account and give it high privileges so they can return later (MITRE T1136.002 and T1098). Each step produces separate low-level log events; the danger is that nobody connects them.

Method. Windows audit policy was enabled on the Domain Controller; the Wazuh agents on the Domain Controller (agent 005) and the Windows 10 workstation (agent 006) forwarded events to the Wazuh manager on an Ubuntu VM. I searched for identity events, built a timeline, triaged the surrounding alerts, wrote a custom correlation rule (ID 100500) and proved it by repeating the attack with a second account.

Verified findings

- The account **svc_backup** was created and enabled by **HOPELABS\Administrator** on 29 Oct 2026 at about 11:01 (local, UTC+1). Wazuh raised rules 60109/60110 (level 8) — E-07, E-08.
- At 11:15:48 the account was added to **Domain Admins** (event 4728). Wazuh rule 60159 fired at **level 12** — E-09 to E-14.
- At 11:39:17 an event 4738 ("user account changed") on svc_backup was recorded under **ANONYMOUS LOGON** with no changed attributes. It remains unexplained and is rated low-to-medium — E-35 to E-39.
- On 31 Oct 2026 at 09:06:02 svc_backup **signed in interactively** (logon type 2) on the Windows 10 workstation DESKTOP-T6T1N2V (event 4624, rule 60118) — E-25 to E-31.
- Two alerts that followed the sign-in (rule 61634 level 12; rule 92052 level 4) were investigated and judged **false positives**: legitimate Windows binaries run by SYSTEM — E-41 to E-48.
- Default Wazuh rules raised the creation and the promotion as **separate** alerts. A custom rule, **100500** (level 13), now links them and fired on 2 Nov 2026 at 12:14:04 for a second test account (svc_test2) — E-49 to E-53.
- A domain sign-in failure on Windows 10 was traced to DNS: the workstation asked the NAT adapter's DNS first. Changing the interface metric fixed it — E-15 to E-24.

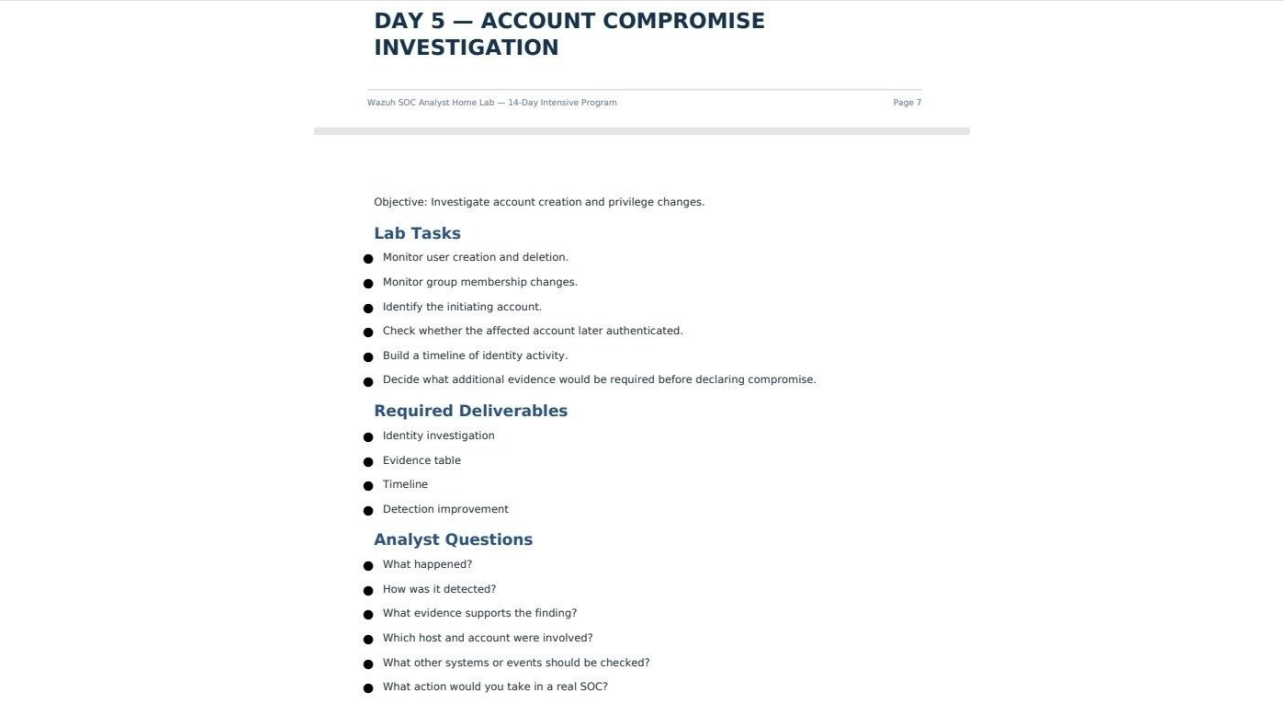
Not verified / not done

- No account deletion (event 4726) was performed, so deletion monitoring is untested.
- Splunk and the Kali VM were part of the lab but produced no evidence used in this report.
- No change ticket or witness exists, so **compromise is not declared**: the lab shows the pattern, not proof of malicious intent.
- The Wazuh version string and the terminal output of the rule syntax check were not captured.

Chapter 2 — Introduction and Objectives

2.1 Background and motivation

A SOC analyst spends most of the day deciding whether an alert is real. Identity events — who created which account, who gave it which rights, and whether it was used — are among the highest-value signals because they reveal persistence and privilege escalation. This lab practices that decision-making on a controlled network.


DAY 5 — ACCOUNT COMPROMISE INVESTIGATION

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Objective: Investigate account creation and privilege changes.

Lab Tasks

- Monitor user creation and deletion.
- Monitor group membership changes.
- Identify the initiating account.
- Check whether the affected account later authenticated.
- Build a timeline of identity activity.
- Decide what additional evidence would be required before declaring compromise.

Required Deliverables

- Identity investigation
- Evidence table
- Timeline
- Detection improvement

Analyst Questions

- What happened?
- How was it detected?
- What evidence supports the finding?
- Which host and account were involved?
- What other systems or events should be checked?
- What action would you take in a real SOC?

Figure 1 (Evidence E-01). The lab brief that defines the objective, tasks, deliverables and analyst questions

Shows: The reference page for this report: objective, six lab tasks, four required deliverables and six analyst questions.
Limitation: It is the course brief, not lab output; its layout was used as the design reference for this report.

2.2 Objectives and measurable success criteria

Table 2. Lab tasks mapped to success criteria and results

Lab task (from the brief)	Success criterion	Result
Monitor user creation and deletion	Creation produces a Wazuh alert; deletion produces a Wazuh alert	Creation: **met** (60109/60110). Deletion: **not tested**
Monitor group membership changes	Promotion to Domain Admins produces an alert	**Met** (4728, rule 60159, level 12)
Identify the initiating account	Initiator named from the event	**Met**: Administrator (the 11:39 event shows ANONYMOUS LOGON, unexplained)
Check whether the affected account later authenticated	A logon event for the new account found	**Met**: 4624 type 2 on Windows 10 (not found on the DC search)
Build a timeline of identity activity	Ordered table with sources	**Met** (Chapter 8, reconstructed timeline)
Decide what extra evidence is needed before declaring compromise	Explicit list	**Met** (Chapter 11)

2.3 Scope

In scope: the Domain Controller (agent 005), the Windows 10 workstation (agent 006) and the Wazuh manager. Out of scope: Kali-based attack simulation, Splunk searches, other hosts, and automated response.

Chapter 3 — Technical Background

Terms are defined in plain language first, then technically.

3.1 Active Directory, domains and Domain Admins

Plain language: a domain is a company-wide user database kept on a Domain Controller. **Technically:** Active Directory stores accounts as objects; every account and group has a SID. The group **Domain Admins** (well-known SID ending in -512) has full control of every machine in the domain, so adding an account to it is the most sensitive identity change possible.

3.2 From event to dashboard: six separate stages

A beginner often treats "the log" as one thing. It is six stages, and a failure at any one makes the alert disappear:

1. **Event generation** — Windows writes an entry to its Security log, but only if the audit policy is enabled (Step 6.2).
2. **Log collection** — the Wazuh agent reads the entry from the Windows Event Channel.
3. **Log forwarding** — the agent sends it over the network to the Wazuh manager.
4. **Event processing** — the manager decodes it into fields (for example `data.win.eventdata.targetUserName`).
5. **Rule evaluation and alert generation** — rules compare the fields; a match becomes an alert with a rule ID and level.
6. **Dashboard display** — the alert is indexed and shown in Threat Hunting, where you search it.

3.3 Windows event IDs used

Table 3. Event IDs referenced in the investigation

Event ID	Meaning	Why it matters here
4720	A user account was created	Start of the persistence pattern (not captured in screenshots)
4722	A user account was enabled	Seen for svc_backup and svc_test2
4724	An attempt was made to reset an account's password	Searched; no result for svc_backup
4726	A user account was deleted	Not exercised
4728	A member was added to a security-enabled *global* group	Domain Admins promotion
4738	A user account was changed	Seen on svc_backup (twice) and baseline noise
4624	An account was successfully logged on	Proves the new account was used
Sysmon 1	Process created (Sysmon)	Used to judge rules 61634 and 92052

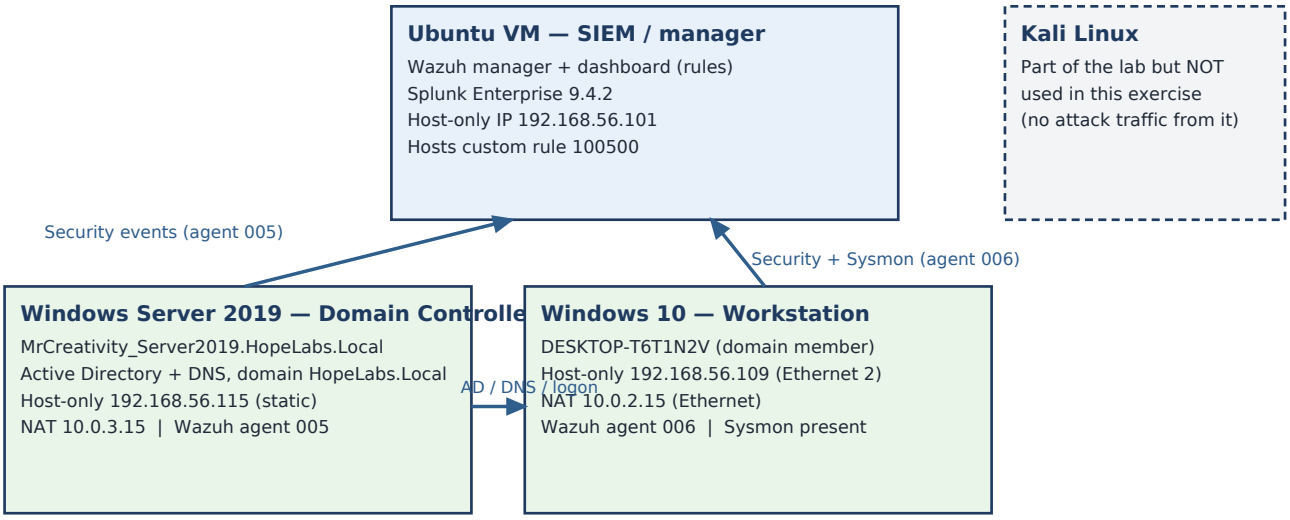
3.4 Logon types, anonymous logon and MITRE techniques

- **Logon type 2** = interactive (someone at the console). Type 3 = network, type 10 = remote desktop.
- **ANONYMOUS LOGON** (SID S-1-5-7, logon ID 0x3E6) is a built-in identity for sessions that supply no credentials. Windows uses it for some background activity, so it is suspicious only in context.
- **T1136.002** Create Account: Domain Account; **T1098** Account Manipulation; **T1078** Valid Accounts — techniques the Wazuh rules map to.
- **DNS and AD:** a domain member finds its Domain Controller through DNS. If DNS is wrong, domain sign-in fails even though the network works (Chapter 10).

Chapter 4 — Lab Environment and Architecture

Table 4. Lab machines and their roles

Machine	Role	Key facts (from evidence)
Ubuntu VM	Wazuh manager and dashboard; Splunk 9.4.2	Dashboard address 192.168.56.101; manager name "Ubuntu"; alert index wazuh-alerts-4.x-YYYY.MM.DD
Windows Server 2019 VM	Domain Controller, AD and DNS	Computer name MrCreativity_Server2019.HopeLabs.Local; domain HopeLabs.Local; static 192.168.56.115 (gateway 192.168.56.1, DNS 192.168.56.115); NAT adapter 10.0.3.15; Wazuh agent name window-server-2019, ID 005
Windows 10 VM	Domain-joined workstation	Host DESKTOP-T6T1N2V; 192.168.56.109 on *Ethernet 2* (host-only); 10.0.2.15 on *Ethernet* (NAT); Wazuh agent ID 006; Sysmon events present
Kali Linux VM	Attacker box (not used)	Available in the lab; no evidence in this report



Host-only network 192.168.56.0/24 carries all lab traffic that matters; NAT adapters are for internet access only.

Figure 2. Laboratory architecture and data flow

Shows: Three VMs on the host-only network 192.168.56.0/24: both Windows machines run Wazuh agents that forward events to the Ubuntu manager; Windows 10 authenticates against the Domain Controller. Kali is shown dashed because it was not used.

How data moves. A Windows action (for example adding a user to a group) is written to the Security log on that machine. The local Wazuh agent forwards it to the manager at 192.168.56.101, rules evaluate it, and the alert appears in the dashboard. Domain sign-ins from Windows 10 are checked by the Domain Controller over the same host-only network.

> **NOTE.** Each Windows VM has two network adapters. The host-only adapter carries lab traffic; the NAT adapter gives internet access. Having two adapters is what caused the DNS problem in Chapter 10.

Chapter 5 — Prerequisites and Safety

5.1 Prerequisites

- A hypervisor (this lab uses Oracle VirtualBox) with three VMs: Ubuntu with Wazuh manager and dashboard, Windows Server 2019 promoted to a Domain Controller for HopeLabs.Local, and Windows 10 joined to that domain.
- Wazuh agents installed, enrolled and showing **Active** on both Windows machines (the lab owner confirmed this before the exercise; the Agents page was not captured).
- Sysmon on Windows 10 is optional for the core lab but was present and used in Step 6.12.
- Credentials: a domain administrator on the Domain Controller (here HOPELABS\Administrator) and a sudo-capable user on Ubuntu.
- Basic comfort opening PowerShell as administrator and a terminal on Ubuntu — everything else is explained.

5.2 Safety and scope

- Use **only** your own isolated lab. Creating Domain Admin accounts on a real network is a serious security event.
- Use throwaway passwords. The commands in this report contain example lab passwords that also appear in some screenshots; never reuse them.
- **Take a snapshot of every VM before you start.** The lab VMs here are named with "SnapShots"; a snapshot is the fastest rollback.
- Back up the Wazuh rules file before editing it (command card C18).
- Clean up afterwards (Chapter 15, Part H). The evidence does not show the test accounts being removed, so they should be considered still present.

> **WARNING.** An administrator who runs these steps is acting as both attacker and defender. Do not run the attack steps on any system you do not own.

Chapter 6 — Step-by-Step Implementation

Follow the steps in order. Each step says **where** to work (which VM), **why**, the **exact command** with every part explained, the **expected** result, what **I actually observed**, and how to **verify**. Commands are numbered C1, C2, …; Wazuh searches are numbered Q1, Q2, …

> **BEGINNER TIP.** PowerShell is not case-sensitive (`get-date` = `Get-Date`), but spaces, hyphens and quotes matter. Most errors in this lab were a missing hyphen or a misspelt command (Chapter 10). Copy and paste whenever possible. 'PS C:\Users\Administrator>' is the prompt — do not type it.

6.1 Step 1 — Start the lab and check it is healthy

Where: VirtualBox, then a browser on the Ubuntu VM. **Why:** if the agents are disconnected, nothing later will show up and you will wrongly think the detection is broken.

1. Start the Ubuntu VM first, then the Domain Controller, then Windows 10. Wait until each reaches its sign-in screen.
2. On Ubuntu open the Wazuh dashboard (in this lab the address is 192.168.56.101) and sign in.
3. Open **Agents** and confirm window-server-2019 (005) and DESKTOP-T6T1N2V (006) show **Active**.

> **EVIDENCE GAP.** The Agents page was not screenshotted. Agent health is shown indirectly: events from both agents appear in later figures.

6.2 Step 2 — Turn on account auditing on the Domain Controller

Where: Domain Controller. **Why:** Windows only records "account created / group changed" events if the matching audit subcategories are switched on. Without this there is nothing for Wazuh to collect (stage 1 in Section 3.2). Open PowerShell as administrator: right-click Start, choose **Windows PowerShell (Admin)**.

C1 Enable auditing of user-account changes

Where / who	**Domain Controller, PowerShell** | Privilege: Administrator (elevated)
Purpose	Make Windows write Security events 4720/4722/4724/4726/4738 for user-account actions.
Command	`auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable`
Breaking it down	**Part** / **What it means** `auditpol` — Built-in Windows tool that edits the *audit policy* (what the system records). `/set` — Change a setting (the opposite of /get). `/subcategory:"User Account Management"` — The exact name of one audit setting. Quotes are required because the name contains spaces. `/success:enable` — Record the event when the action succeeds. `/failure:enable` — Also record it when the action fails (useful for spotting guessing attacks).
Expected result	The line "The command was successfully executed."
Observed in this lab	Both set commands printed "The command was successfully executed." (Figure 3).
How to verify	Run C3 and confirm the subcategory shows **Success and Failure**.
If it fails	"Access is denied" or an error about privileges → you did not open PowerShell as administrator; close it and reopen with **Run as administrator**.

C2 Enable auditing of group-membership changes

Where / who	**Domain Controller, PowerShell** | Privilege: Administrator (elevated)
Purpose	Make Windows write events such as 4728 when someone is added to a security group.
Command	`auditpol /set /subcategory:"Security Group Management" /success:enable /failure:enable`

C2 Enable auditing of group-membership changes	
Breaking it down	**Part** / **What it means** `/subcategory:"Security Group Management"` — The audit setting for groups. Event 4728 (member added to a global group) is controlled by it. `other parts` — Same meaning as in C1.
Expected result	The line "The command was successfully executed."
Observed in this lab	Printed successfully (Figure 3).
How to verify	Run C3.
If it fails	Same as C1.

C3 Display the current audit policy for the Account Management category	
Where / who	**Domain Controller, PowerShell** | Privilege: Administrator (elevated)
Purpose	Confirm that C1 and C2 really changed the policy.
Command	`auditpol /get /category:"Account Management"`
Breaking it down	**Part** / **What it means** `/get` — Show the current policy instead of changing it. `/category:"Account Management"` — A *category* is a group of subcategories. Use the category name here, not the subcategory name.
Expected result	A table listing six subcategories (Computer Account, Security Group, Distribution Group, Application Group, Other Account Management Events, User Account Management), each set to Success and Failure.
Observed in this lab	The table showed all six at "Success and Failure". An earlier attempt with /category:"User Account Management" returned "Error 0x00000057 occurred: The parameter is incorrect" because that name is a *subcategory*, not a category (Figure 3).
How to verify	Read the Setting column for the two subcategories you changed.
If it fails	Error 0x00000057 → a name is wrong. Use `auditpol /list /category` to see valid category names.

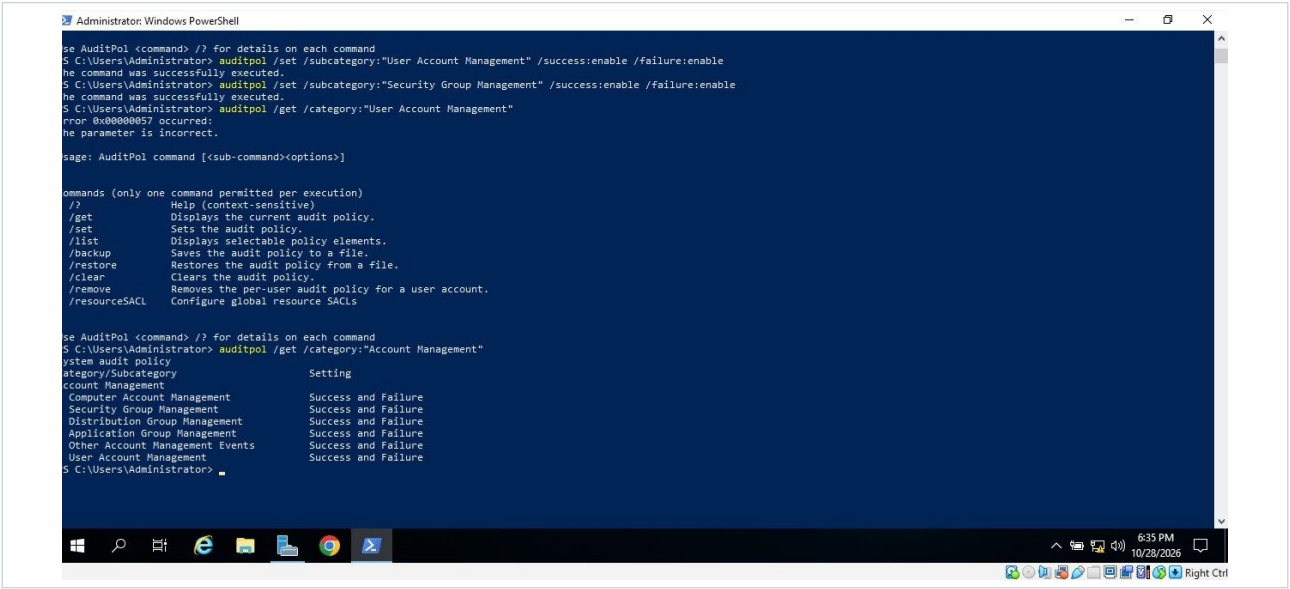


Figure 3 (Evidence E-03). Audit policy enabled on the Domain Controller (C1–C3)

Shows: The two successful auditpol /set commands, the 0x00000057 error from the wrong category name, and the final table with all Account Management subcategories at Success and Failure. **Limitation:** Taken from the PowerShell console; it proves the policy at that moment, not that events were later collected.

6.3 Step 3 — Confirm Domain Controller events reach Wazuh

Where: Wazuh dashboard. **Why:** this checks stages 2–6 of Section 3.2 before you rely on the lab. In the dashboard go to **Threat Hunting**, choose the agent **window-server-2019**, then open the **Events** tab. The agent filter chip (agent.id: 005) appears under the search bar.

Q1	`data.win.system.channel: Security`
Meaning	Show only events whose Windows channel is "Security" (the log that holds logons and account changes). Syntax is `field: value`.
Used to	prove that Security-log events from the Domain Controller reach Wazuh.

Set the time range (top right) wide enough — for example **Last 7 days** — then press **Refresh**. Beginners often forget Refresh and read a stale result.

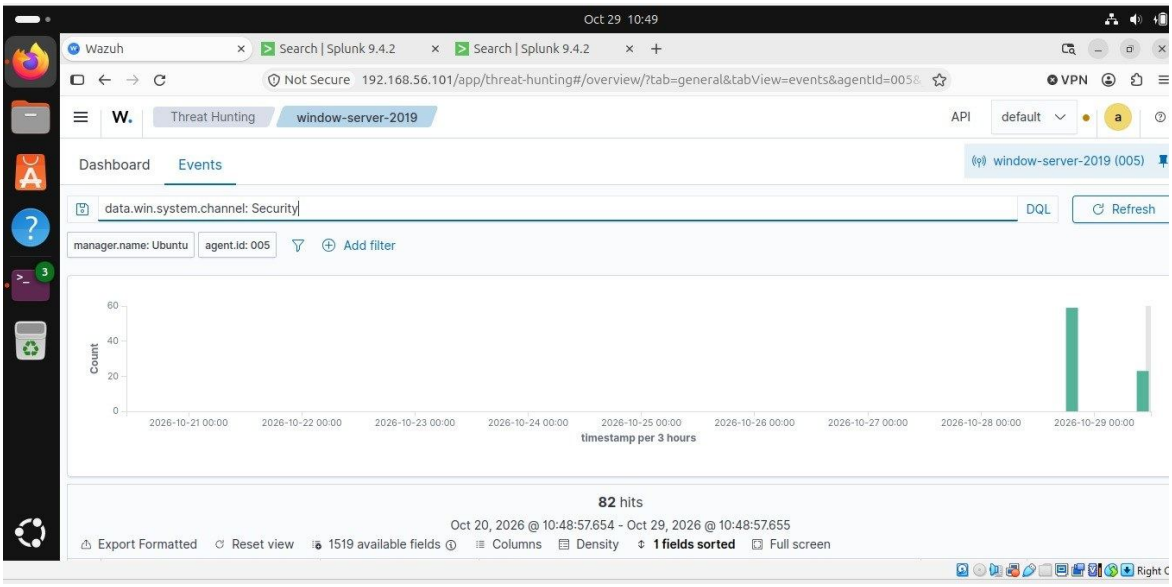


Figure 4 (Evidence E-04). Domain Controller Security-channel events in Wazuh (Q1)
Shows: The search data.win.system.channel: Security on agent 005 returned 82 hits with a visible spike on 29 October. **Limitation:** Read at the time of the first check; the count grew afterwards.

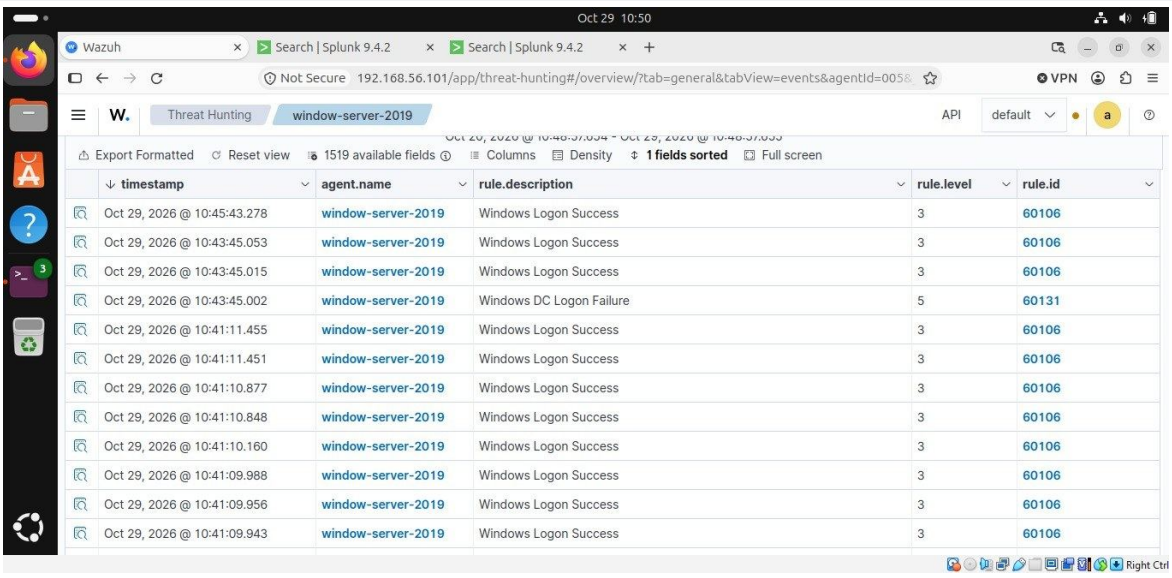


Figure 5 (Evidence E-05). Sample of the Security alerts: Windows Logon Success (60106) and DC Logon Failure (60131)
Shows: Alert rows from window-server-2019 with rule descriptions, levels (3 and 5) and rule IDs. **Limitation:** These are routine logons, not the account activity yet.

6.4 Step 4 — Create the rogue account (the 'attack')

Where: Domain Controller, administrator PowerShell. **Why:** creating a new account is the first half of the persistence pattern. We record the time first so we can match it to the logs.

C4 Show the current date and time

Where / who	**Domain Controller, PowerShell** | Privilege: Any user
Purpose	Write down a time marker for the timeline.
Command	`Get-Date`
Breaking it down	**Part** / **What it means** `Get-Date` — A PowerShell *cmdlet* (built-in command) that prints the system clock. No options are needed.
Expected result	One line such as "Thursday, October 29, 2026 10:53:56 AM".
Observed in this lab	It printed Thursday, October 29, 2026 10:53:56 AM. The account was created later; the taskbar clock read 11:02 AM when it was verified (Figure 6).
How to verify	Compare the printed time with the taskbar clock.
If it fails	Nothing normally fails. Remember the clock is in the VM's local time (UTC+1 here), while Windows events store UTC.

C5 Create the domain user svc_backup

Where / who	**Domain Controller, PowerShell** | Privilege: Domain administrator
Purpose	Create a new, enabled Active Directory account. The name svc_backup looks like a normal service account, which is exactly why attackers choose such names.
Command	`New-ADUser -Name "svc_backup" ` `-SamAccountName svc_backup ` `-AccountPassword (ConvertTo-SecureString "P@ssw0rd123!" -AsPlainText -Force) ` `-Enabled $true`
Breaking it down	**Part** / **What it means** `New-ADUser` — Cmdlet from the ActiveDirectory module that creates a user object. `-Name "svc_backup"` — The object's display name in Active Directory. `-SamAccountName svc_backup` — The sign-in name; with the domain it becomes HOPELABS\svc_backup. `-AccountPassword (...)` — The initial password. AD requires a *secure string*, not plain text. `ConvertTo-SecureString "..." -AsPlainText -Force` — Turns text into a secure string. -AsPlainText says you are supplying readable text; -Force confirms you accept that. `( ... )` — Parentheses make PowerShell run the inner part first and pass the result on. `-Enabled $true` — Account is active. `$true` is PowerShell's yes/true value. `` ` (backtick at line end) `` — Continues the command on the next line. You can type everything on one line instead.
Expected result	No output at all. In PowerShell, silence means success for this cmdlet.
Observed in this lab	No output; the next command (C6) found the account. I typed it on one line and used the shortened parameter -Enable, which PowerShell accepts when unambiguous (Figure 6).
How to verify	Run C6.
If it fails	"The password does not meet the length, complexity, or history requirement" → use a longer password with upper, lower, digits and symbols. "The specified account already exists" → choose another name. "The term 'New-ADUser' is not recognized" → you are not on the Domain Controller (or run `Import-Module ActiveDirectory`). "Access is denied" → reopen as administrator.
Note	Security: the password stays in the PowerShell history in clear text. Acceptable only for a disposable lab password.

C6 Read back the new account	
Where / who	**Domain Controller, PowerShell** | Privilege: Domain administrator
Purpose	Prove the account exists and is enabled.
Command	`Get-ADUser svc_backup`
Breaking it down	**Part** / **What it means** `Get-ADUser` — Cmdlet that reads an account from Active Directory. `svc_backup` — The identity to look up (positional value of -Identity).
Expected result	A block of properties: DistinguishedName, Enabled, SamAccountName, SID and others.
Observed in this lab	Enabled: True; DistinguishedName CN=svc_backup,CN=Users,DC=HopeLabs,DC=Local; SID ending -1129. My first attempt, typed as get--aduser (two hyphens), failed with CommandNotFoundException (Figure 6).
How to verify	Check Enabled is True and note the last number of the SID (-1129): it is how we later recognise this account.
If it fails	"Cannot find an object with identity" → C5 did not run or the name is misspelt.

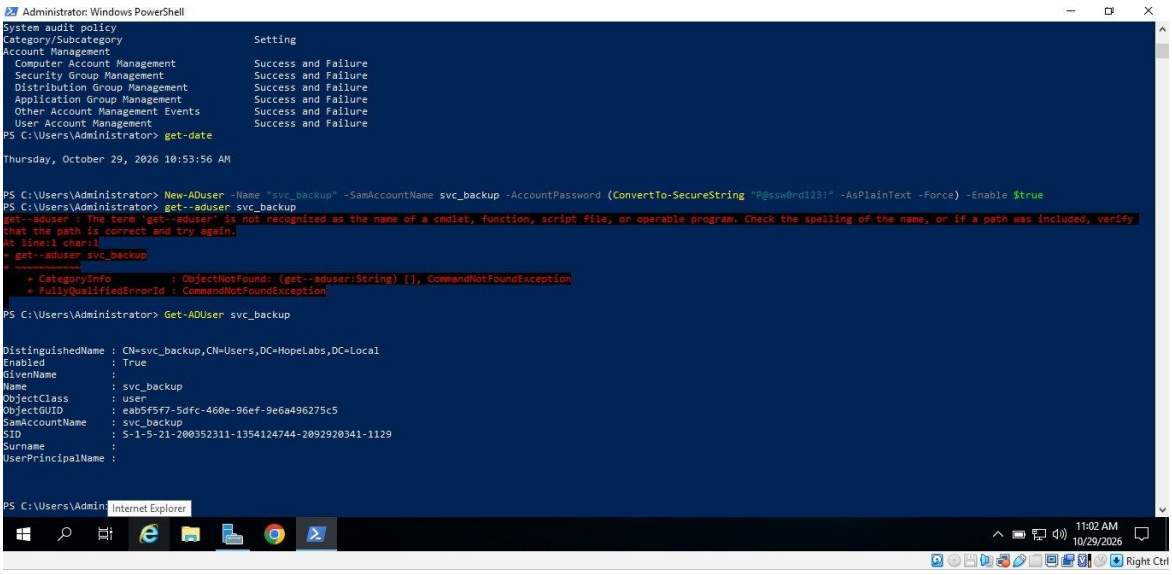


Figure 6 (Evidence E-06). Creation and verification of svc_backup (C4–C6)

Shows: Get-Date, New-ADUser, the typo error get--aduser, then Get-ADUser showing the account enabled with SID ending -1129. **Limitation:** The screenshot shows the commands, not the Wazuh result; that comes next.

6.5 Step 5 — Find the creation in Wazuh

Where: Wazuh dashboard, agent window-server-2019. Wait up to a minute: Wazuh showed events roughly one minute after Windows wrote them. Set the range to cover 29 October.

Q2	`data.win.system.eventID: (4720 or 4722 or 4724 or 4726 or 4728 or 4732 or 4738 or 4756)`
Meaning	Match any of these identity event IDs. `or` must be lower-case or upper-case *letters* — a zero (`0r`) breaks the query, which happened once in this lab.
Used to	list every account and group change in one view.

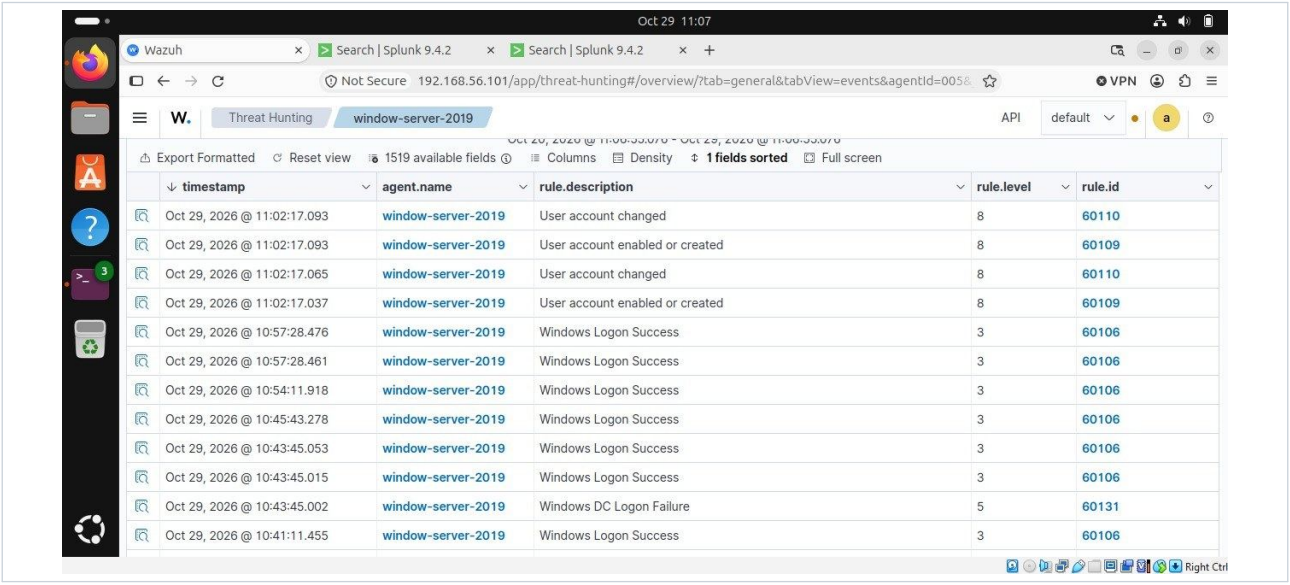


Figure 7 (Evidence E-07). First alerts for svc_backup at about 11:02 on 29 October
Shows: Rows with rule 60109 "User account enabled or created" and rule 60110 "User account changed", both level 8, on window-server-2019. **Limitation:** The list does not show the account name; open a row for that (next figure).

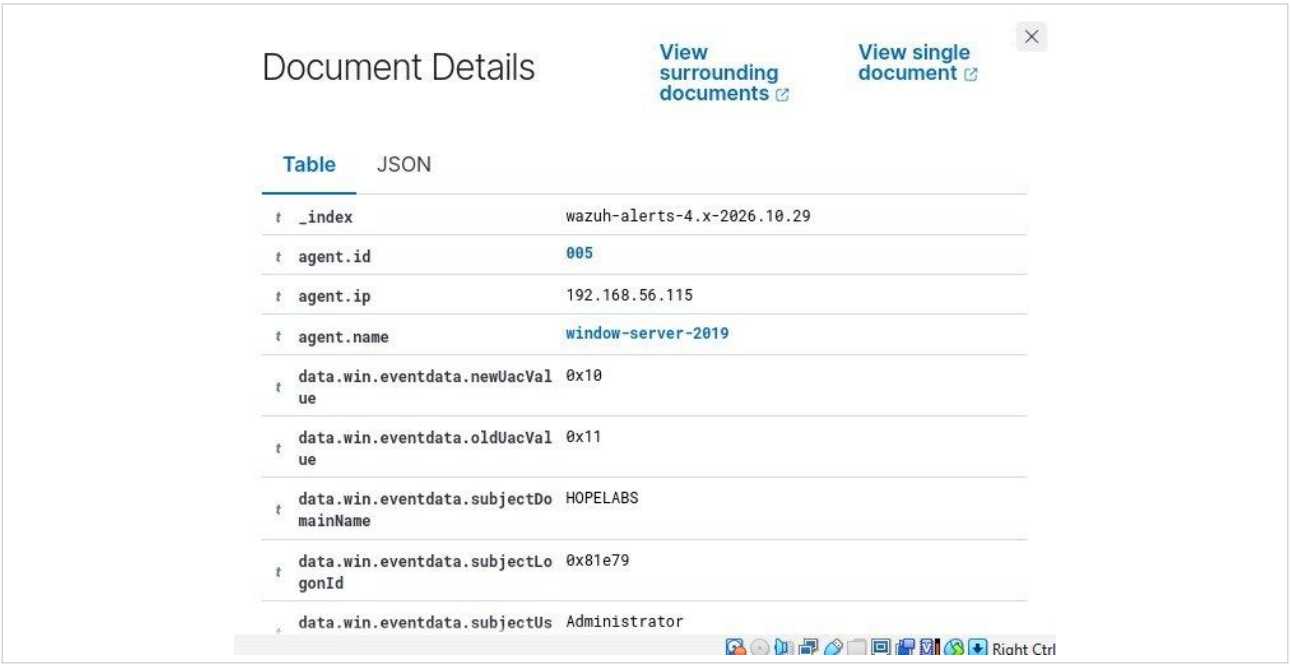


Figure 8 (Evidence E-08). Details of the account-change alert (event 4738)
Shows: Agent 005, 192.168.56.115; data.win.eventdata.oldUacValue 0x11 and newUacValue 0x10; subject HOPELABS, logon ID 0x81e79, user Administrator. **Limitation:** The exact meaning of the 0x11 → 0x10 flag change was not decoded; it coincides with the 4722 "account was enabled" event and is treated as part of enabling the account.

How to read it. `subjectUserName` is the account that *did* the action (Administrator); `targetUserName` is the account it was done *to* (svc_backup). The logon ID 0x81e79 identifies the administrator's session and appears again on the group change, linking both actions to one session. Rule 60109/60110 are mapped to MITRE T1098 (Account Manipulation, Persistence).

> **EVIDENCE GAP.** An event 4720 ("account created") record was not captured in the screenshots; creation is evidenced by rule 60109 ("enabled or created") and event 4722.

6.6 Step 6 — Promote the account to Domain Admins

Where: Domain Controller, administrator PowerShell. **Why:** this is the escalation step: from an ordinary account to one that controls the whole domain.

C7 Add svc_backup to Domain Admins	
Where / who	**Domain Controller, PowerShell** | Privilege: Domain administrator
Purpose	Grant the new account full administrative control of the domain.
Command	`Add-ADGroupMember -Identity "Domain Admins" -Members svc_backup`
Breaking it down	**Part** / **What it means** `Add-ADGroupMember` — Cmdlet that adds accounts to an AD group. `-Identity "Domain Admins"` — The group to change. Quotes are needed because of the space. `-Members svc_backup` — The account(s) to add; several can be listed separated by commas.
Expected result	No output (silence = success).
Observed in this lab	No output; C8 then listed the account (Figure 9). Performed at 11:14:50 AM per Get-Date.
How to verify	Run C8.
If it fails	"Cannot find an object with identity: 'Domain Admins'" → the group name is misspelt. "Access is denied" → not elevated.
Note	Security: Domain Admins members can read, change or delete almost anything in the domain.

C8 List the members of Domain Admins	
Where / who	**Domain Controller, PowerShell** | Privilege: Domain administrator
Purpose	Verify the promotion.
Command	`Get-ADGroupMember "Domain Admins" | Select Name`
Breaking it down	**Part** / **What it means** `Get-ADGroupMember` — Lists the members of a group. `|` — Pipe: sends the output of the left command to the right command. `Select Name` — Short for Select-Object; keep only the Name column.
Expected result	A list with the built-in Administrator and the new account.
Observed in this lab	Administrator and svc_backup were listed (Figure 9).
How to verify	Look for svc_backup in the list.
If it fails	An empty or error output means C7 failed; repeat it.

```
PS C:\Users\Administrator> Get-ADUser svc_backup


DistinguishedName : CN=svc_backup,CN=Users,DC=HopeLabs,DC=Local
Enabled           : True
GivenName         :
Name              : svc_backup
ObjectClass       : user
ObjectGUID        : eab5f5f7-5dfc-460e-96ef-9e6a496275c5
SamAccountName    : svc_backup
SID               : S-1-5-21-200352311-1354124744-2092920341-1129
Surname           :
UserPrincipalName :



PS C:\Users\Administrator> get-date

Thursday, October 29, 2026 11:14:50 AM


PS C:\Users\Administrator> Add-ADGroupMember -Identity "Domain Admins" -Members svc_backup
PS C:\Users\Administrator> Get-ADGroupMember "Domain Admins" | Select Name

Name
----
Administrator
svc_backup


PS C:\Users\Administrator>
```

Figure 9 (Evidence E-09). svc_backup added to Domain Admins (C4, C7, C8)

Shows: Get-ADUser output, a Get-Date of 11:14:50 AM, Add-ADGroupMember and the group listing showing Administrator and svc_backup.

6.7 Step 7 — Find the group change in Wazuh

Q3	`data.win.system.eventID: 4728`
Meaning	Show only "member added to a security-enabled global group" events.
Used to	find the Domain Admins promotion.

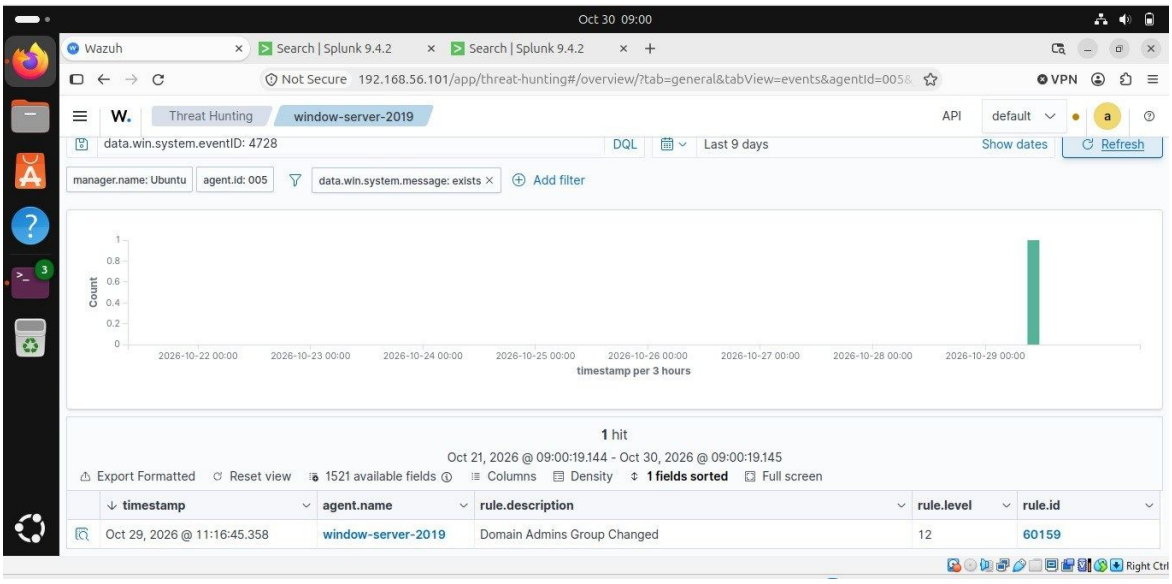


Figure 10 (Evidence E-10). Wazuh alert for the group change — "Domain Admins Group Changed", level 12

Shows: One hit on 29 October at 11:16:45 with rule ID 60159 and level 12. **Limitation:** Index time 11:16:45 is about a minute later than the event itself (E-14).

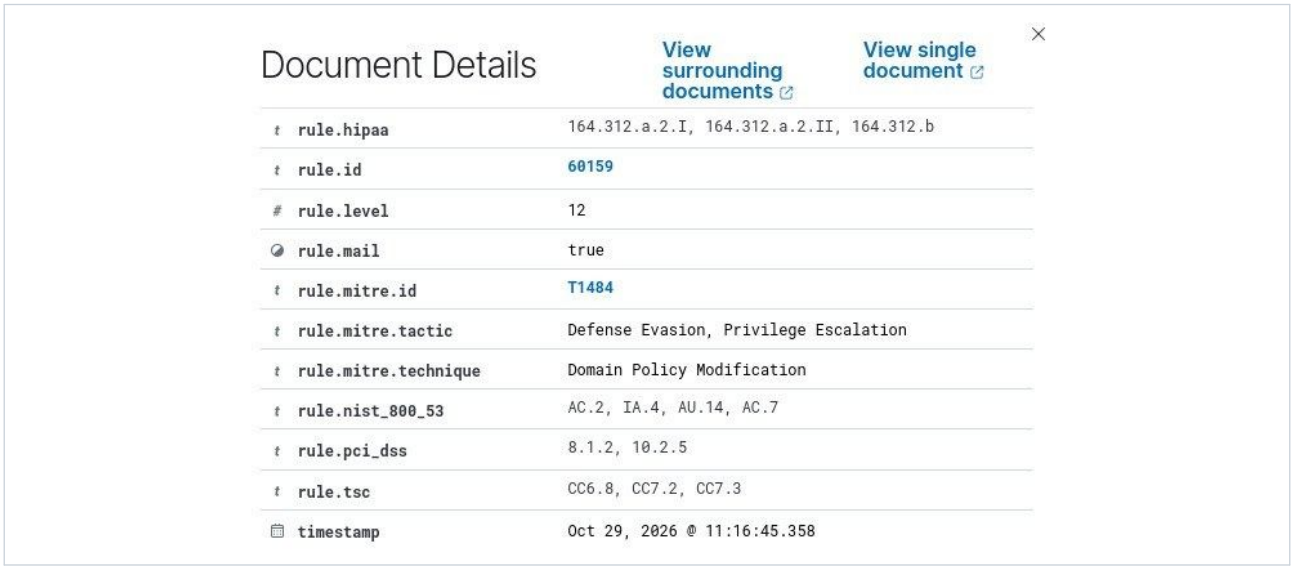


Figure 11 (Evidence E-11). Rule 60159 details

Shows: Rule level 12, MITRE T1484 (Domain Policy Modification: Defense Evasion, Privilege Escalation), timestamp 29 Oct 2026 11:16:45.358. **Limitation:** The rule's MITRE mapping (T1484) differs from the T1098 mapping of the account rules; both are Wazuh's own labels.

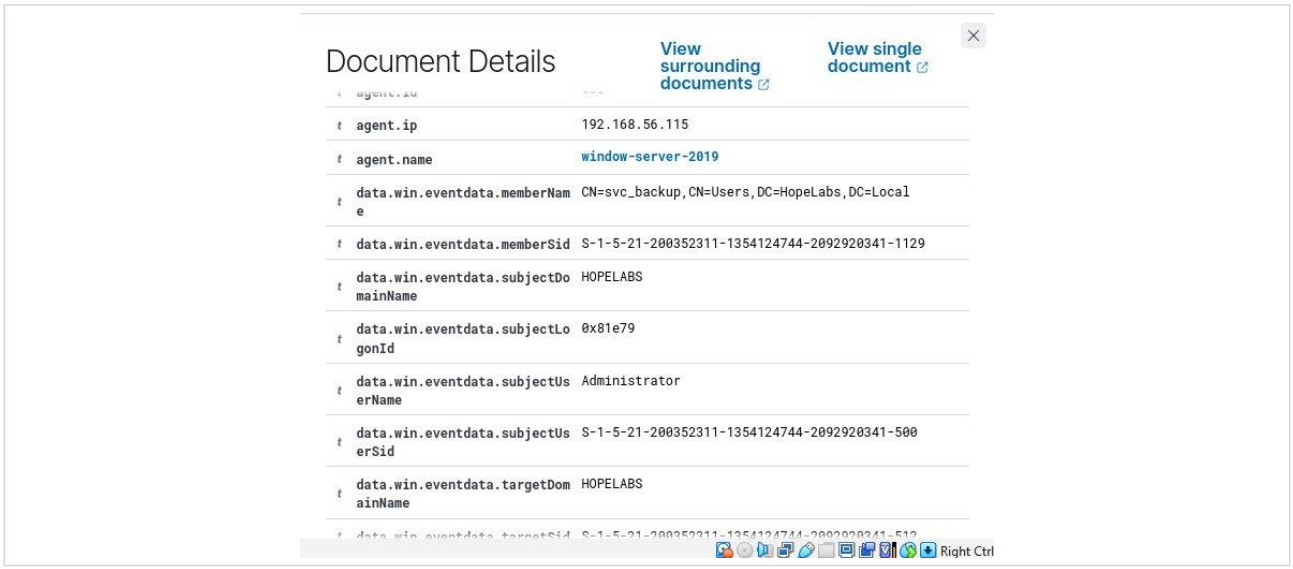


Figure 12 (Evidence E-12). Who was added, and by whom

Shows: data.win.eventdata.memberName CN=svc_backup,CN=Users,DC=HopeLabs,DC=Local; member SID ending -1129; subject Administrator (HOPELABS), logon ID 0x81e79.

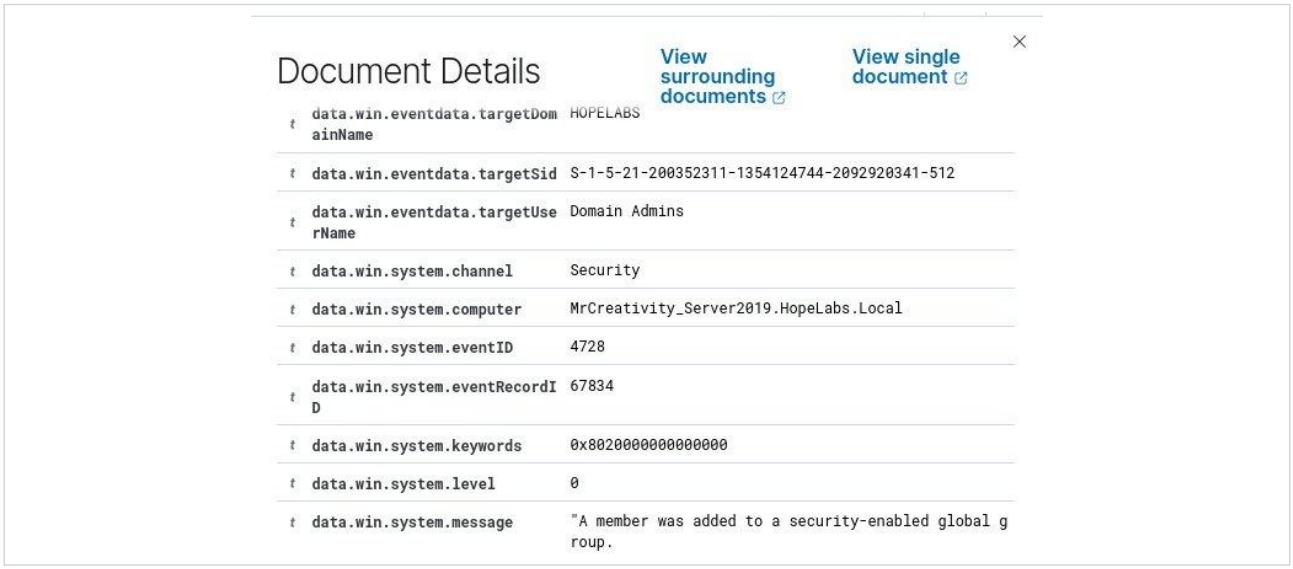


Figure 13 (Evidence E-13). Which group changed

Shows: targetUserName Domain Admins, target SID ending -512, event ID 4728, record 67834 on MrCreativity_Server2019.HopeLabs.Local.

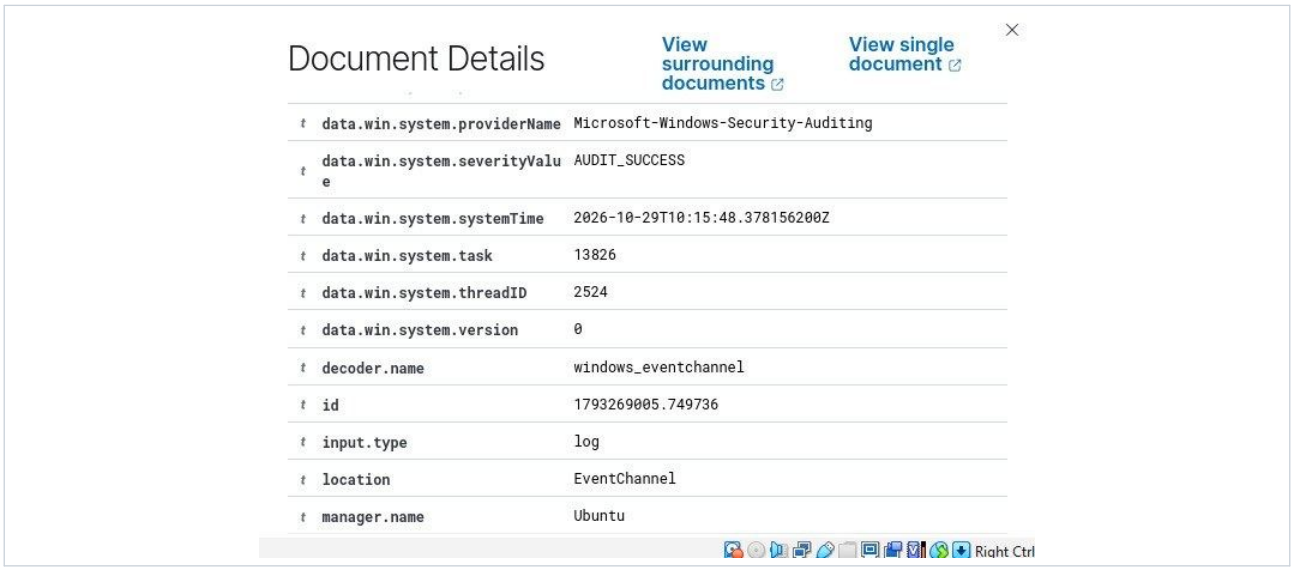


Figure 14 (Evidence E-14). The event's own timestamp

Shows: systemTime 2026-10-29T10:15:48Z = 11:15:48 local (UTC+1), Microsoft-Windows-Security-Auditing, AUDIT_SUCCESS. **Limitation:** Use this time, not the dashboard time, for the timeline.

Finding: the same administrator session (0x81e79) created the account and promoted it about 14 minutes later. Wazuh scored the creation at level 8 and the promotion at level 12, but as separate alerts.

6.8 Step 8 — Windows 10 cannot sign in as the new account (troubleshooting)

Where: Windows 10. On the sign-in screen choose **Other user** and enter `H0PELABS\svc_backup` with the lab password. It failed. The root cause analysis is in Chapter 10; this section gives the commands in order so you can reproduce the fix.

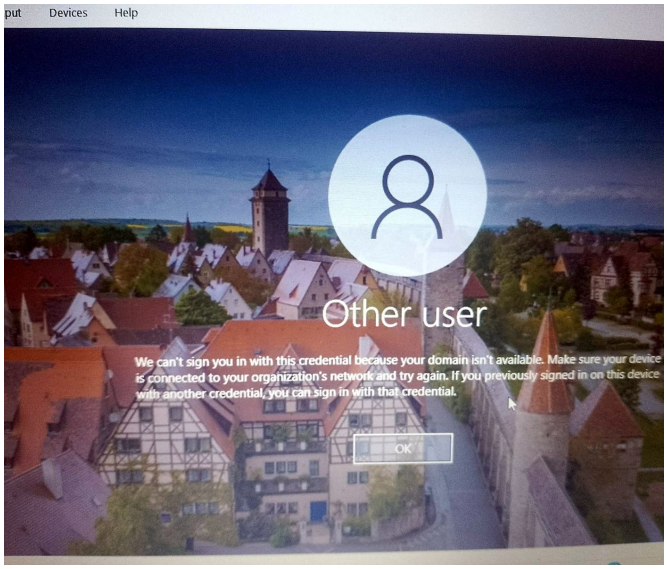


Figure 15 (Evidence E-15). Sign-in failure on Windows 10
Shows: Message: "We can't sign you in with this credential because your domain isn't available. Make sure your device is connected to your organization's network and try again." **Limitation:** A photo of the screen, auto-cropped and contrast-enhanced.

C9 Show the network configuration	
Where / who	**Windows 10 (or the DC), PowerShell** | Privilege: Any user
Purpose	See the IP address, DNS servers, gateway and DHCP state of every adapter.
Command	`ipconfig /all`
Breaking it down	**Part** / **What it means** `ipconfig` — Windows tool that prints network settings. `/all` — Include details such as DNS servers, DHCP and MAC addresses (without it you only get addresses).
Expected result	One block per adapter. The host-only adapter should show 192.168.56.x and DNS Servers 192.168.56.115.
Observed in this lab	On Windows 10 the host-only adapter listed only placeholder DNS entries (fec0:0:0:ffff::1–3) while the NAT adapter used 10.0.2.3 (Figure 16). On the DC, Ethernet = 192.168.56.115 (static, DNS 192.168.56.115) and Ethernet 2 = 10.0.3.15 via DHCP (Figure 19).
How to verify	Check that DNS Servers on the lab adapter is the Domain Controller.
If it fails	If the output scrolls away, run `ipconfig /all | more` and press Space.

C10 Test basic connectivity to the Domain Controller	
Where / who	**Windows 10, PowerShell** | Privilege: Any user
Purpose	Prove the network path works, so the problem must be higher up (DNS).
Command	`ping 192.168.56.115`
Breaking it down	**Part** / **What it means** `ping` — Sends small test packets (ICMP echo) and waits for replies. `192.168.56.115` — The Domain Controller's host-only address.
Expected result	Four "Reply from 192.168.56.115" lines.

C10 Test basic connectivity to the Domain Controller	
Observed in this lab	I confirmed that the ping succeeded; this was reported during the session and was not captured as a screenshot.
How to verify	Look for replies and a low time.
If it fails	"Request timed out" → wrong IP, VM off, or host-only adapters on different virtual networks.

C11 Ask DNS for the domain name	
Where / who	**Windows 10, PowerShell** | Privilege: Any user
Purpose	Check that the workstation can find the domain through DNS.
Command	`nslookup HopeLabs.Local`
Breaking it down	**Part** / **What it means** `nslookup` — Queries a DNS server and prints the answer and which server answered. `HopeLabs.Local` — The name to look up. Case does not matter. `optional 2nd value` — You can add a DNS server, e.g. `nslookup HopeLabs.Local 192.168.56.115`, to force which server is asked.
Expected result	"Server:" shows the Domain Controller (192.168.56.115) and "Addresses:" lists its IP(s).
Observed in this lab	First: Server UnKnown, Address 10.0.2.3, "Non-existent domain" (Figure 16) — the wrong DNS server was asked. Forcing the DC with the second value worked (Figure 21): Server fileserver.hopelabs.local, 192.168.56.115, with addresses fd17:625c:f037:3:cf0f:d26f:729f:d3ff, 192.168.56.115 and 10.0.3.15.
How to verify	Compare "Server" with the DC's IP.
If it fails	"DNS request timed out" → the server asked is unreachable. "Non-existent domain" → the server asked does not know the zone.

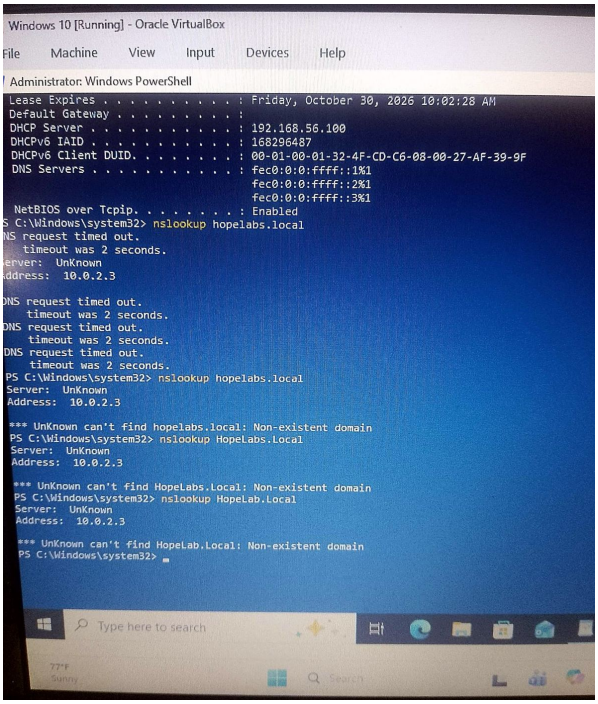


Figure 16 (Evidence E-16). DNS lookups failing on Windows 10

Shows: nslookup hopelabs.local answered by UnKnown 10.0.2.3 with timeouts and "Non-existent domain". **Limitation:** Photo of the screen; text is small but legible at full zoom.

C12 List the network adapters	
Where / who	**Windows 10, PowerShell** | Privilege: Any user

C12 List the network adapters	
Purpose	Learn the exact adapter names you must type in later commands.
Command	`Get-NetAdapter`
Breaking it down	**Part** / **What it means** `Get-NetAdapter` — Lists the physical and virtual network cards with name, status and speed.
Expected result	A table with Name, InterfaceDescription, ifIndex and Status.
Observed in this lab	Two adapters, both Up: Ethernet (ifIndex 6) and Ethernet 2 (ifIndex 3).
How to verify	Both should say Up.
If it fails	No rows → run PowerShell as administrator.

C13 Show which adapter holds which IPv4 address	
Where / who	**Windows 10, PowerShell** | Privilege: Any user
Purpose	Identify the host-only adapter (the 192.168.56.x one).
Command	`Get-NetIPAddress -AddressFamily IPv4 | Select InterfaceAlias, IPAddress`
Breaking it down	**Part** / **What it means** `Get-NetIPAddress` — Lists every IP address on the computer. `-AddressFamily IPv4` — Only the older four-number addresses. `| Select InterfaceAlias, IPAddress` — Keep only the adapter name and the address.
Expected result	One line per adapter.
Observed in this lab	Ethernet 2 = 192.168.56.109 (host-only); Ethernet = 10.0.2.15 (NAT); loopback 127.0.0.1 (Figure 17).
How to verify	The 192.168.56.x line tells you the adapter name to use next.
If it fails	Misreading which adapter is which is the most common error — always check before changing DNS.

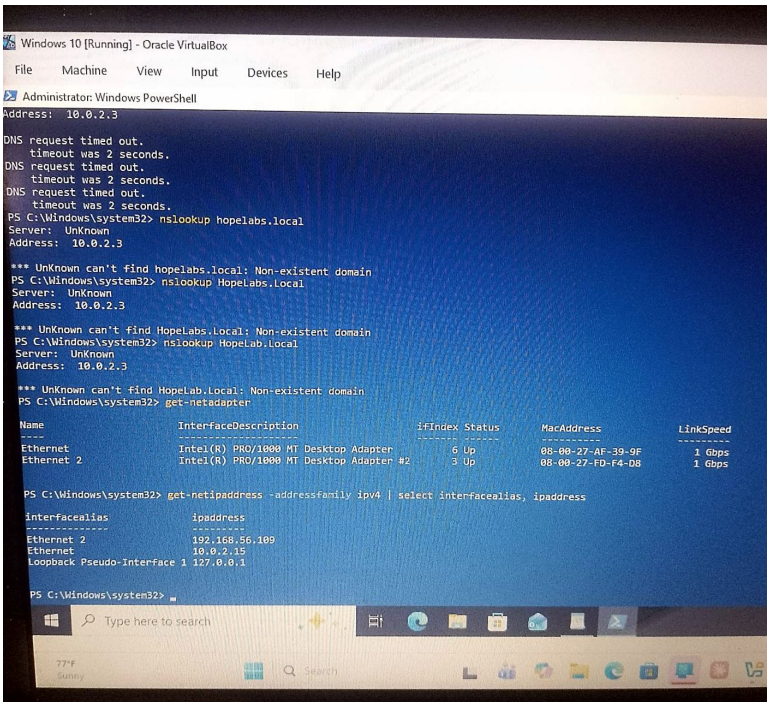


Figure 17 (Evidence E-17). Adapters and addresses on Windows 10 (C12, C13)

Shows: Get-NetAdapter output and the IPv4 listing: Ethernet 2 = 192.168.56.109, Ethernet = 10.0.2.15. **Limitation:** Photo of the screen.

C14 Point the host-only adapter at the Domain Controller for DNS	
Where / who	**Windows 10, PowerShell** | Privilege: Administrator (elevated)
Purpose	Tell Windows to use the DC as DNS server on the adapter that can reach it.
Command	`Set-DnsClientServerAddress -InterfaceAlias "Ethernet 2" ` `-ServerAddresses 192.168.56.115`
Breaking it down	**Part** / **What it means** `Set-DnsClientServerAddress` — Cmdlet that sets which DNS servers an adapter uses. `-InterfaceAlias "Ethernet 2"` — Selects the adapter by name (from C12/C13). `-ServerAddresses 192.168.56.115` — The DNS server list; here only the DC.
Expected result	No output.
Observed in this lab	First attempt wrote serveraddress without the leading hyphen and failed with "A positional parameter cannot be found that accepts argument 'serveraddress'". The corrected command ran silently (Figure 18).
How to verify	Run `Get-DnsClientServerAddress -InterfaceAlias "Ethernet 2"` to see the saved value.
If it fails	"A positional parameter cannot be found" → a hyphen is missing before a parameter name.

C15 Clear the DNS cache	
Where / who	**Windows 10, PowerShell** | Privilege: Administrator
Purpose	Remove old (wrong) answers that Windows remembered.
Command	`ipconfig /flushdns`
Breaking it down	**Part** / **What it means** `/flushdns` — Empties the local DNS resolver cache.
Expected result	The line "Successfully flushed the DNS Resolver Cache."
Observed in this lab	That line appeared (Figure 18).

C15 Clear the DNS cache	
How to verify	Re-run C11.
If it fails	Requires administrator rights.

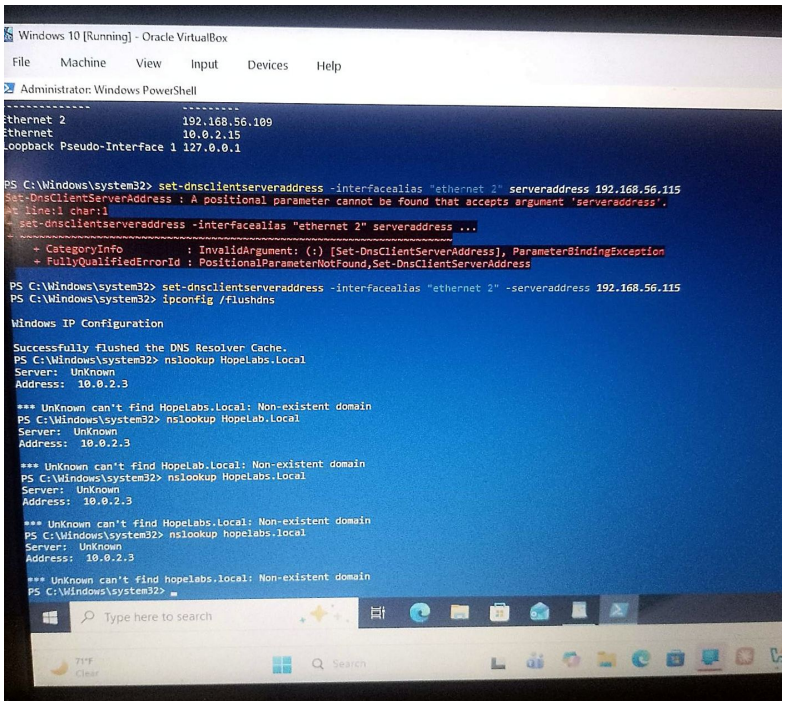


Figure 18 (Evidence E-18). Setting the DNS server and flushing the cache (C14, C15)

Shows: The failed first attempt (missing hyphen), the corrected command, ipconfig /flushdns success, and lookups that still reached 10.0.2.3. **Limitation:** After this change the lookups still reported 10.0.2.3, which led to the metric fix. Photo of the screen.

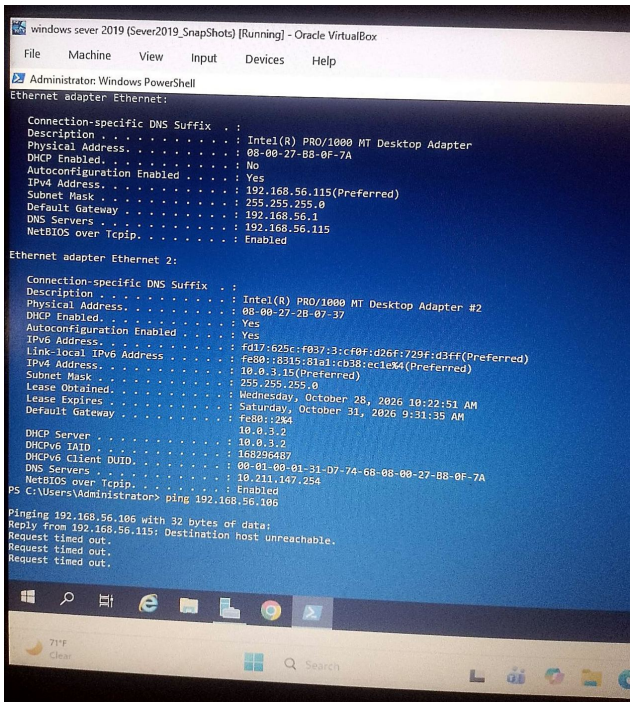


Figure 19 (Evidence E-19). Domain Controller network configuration (C9 on the DC)

Shows: Ethernet 192.168.56.115 (static, gateway 192.168.56.1, DNS 192.168.56.115) and Ethernet 2 10.0.3.15 (DHCP, DNS 10.211.147.254). **Limitation:** The same screen also shows an unrelated ping to 192.168.56.106 (75% loss); the host at that address was not identified.

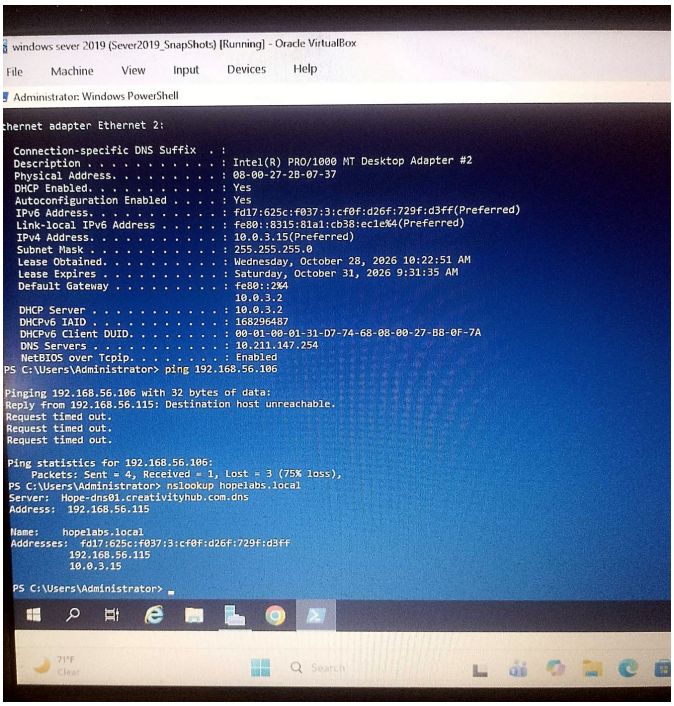


Figure 20 (Evidence E-20). DNS on the Domain Controller itself

Shows: nslookup hopelabs.local on the DC returns 192.168.56.115 and the NAT address 10.0.3.15. **Limitation:** The DC registers its NAT address as well, a multi-homed side effect noted in Chapter 11.

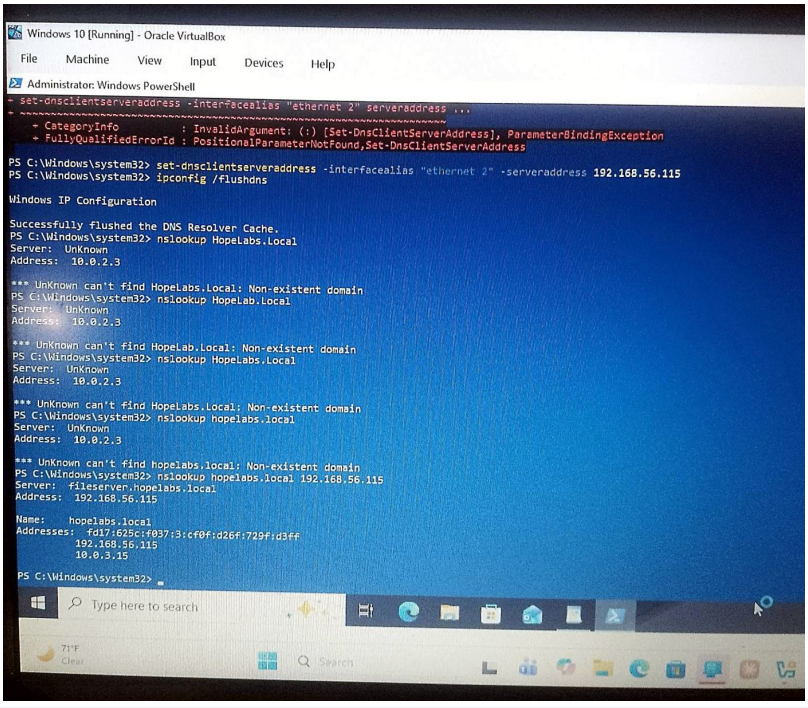


Figure 21 (Evidence E-21). Windows 10 asking the DC directly

Shows: nslookup hopelabs.local 192.168.56.115 succeeds (Server fileserver.hopelabs.local, 192.168.56.115). The same screen shows the earlier failing lookups. **Limitation:** Proves the DC's DNS works and that the workstation was simply asking the wrong server.

C16 Make Windows prefer the host-only adapter	
Where / who	**Windows 10, PowerShell** | Privilege: Administrator (elevated)
Purpose	Change adapter priority so DNS queries go to the Domain Controller first.
Command	`Set-NetIPInterface -InterfaceAlias "Ethernet 2" -InterfaceMetric 5` `Set-NetIPInterface -InterfaceAlias "Ethernet" -InterfaceMetric 100`

C16 Make Windows prefer the host-only adapter

Breaking it down

Part	What it means
`Set-NetIPInterface`	Cmdlet that edits IP settings of an adapter.
`-InterfaceMetric 5`	A cost number: **lower = preferred**. Windows tries the lowest-metric adapter's DNS first.
`-InterfaceMetric 100`	A high number so the NAT adapter is used last.

Expected result	No output.
Observed in this lab	My first tries were misspelt (set-netinterface, set-neiptinterface) or lacked hyphens (Figure 22). The corrected commands ran silently. Afterwards nslookup returned Server Hope-dns01.creativityhub.com.dns, Address 192.168.56.115 (Figure 23).
How to verify	Run C15 and then C11.
If it fails	"The term ... is not recognized" → cmdlet misspelt; copy it exactly.
Note	The server name printed by nslookup differed between runs (fileserver.hopelabs.local vs Hope-dns01.creativityhub.com.dns); both are the same IP. It is the reverse-DNS name for 192.168.56.115 and the cause of the difference was not investigated.

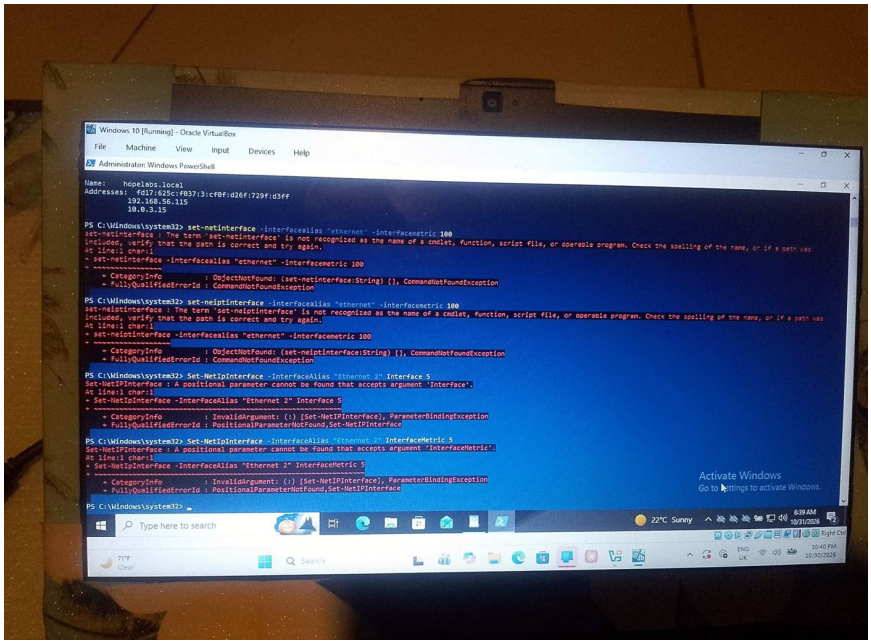


Figure 22 (Evidence E-22). Typing errors while setting the metric

Shows: CommandNotFoundException for set-netinterface and set-neiptinterface and parameter-binding errors for missing hyphens. **Limitation:** Errors left in on purpose; they are real troubleshooting evidence. Photo of the screen.

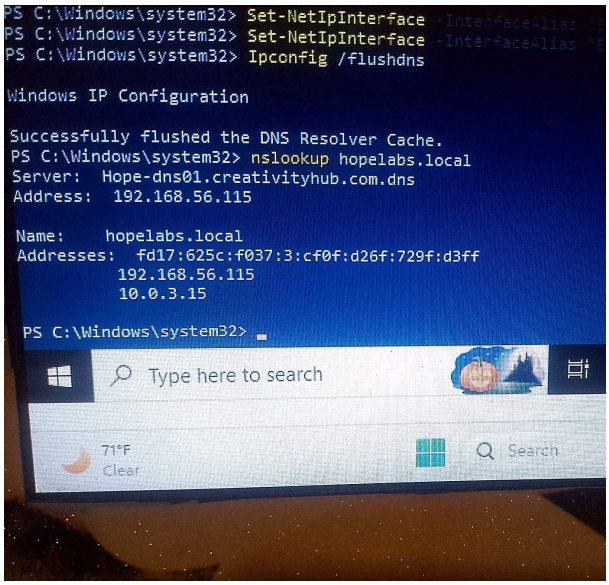


Figure 23 (Evidence E-23). DNS now answered by the Domain Controller

Shows: The corrected Set-NetIPInterface lines (metric 5 and 100), flushdns success and nslookup hopelabs.local answered by 192.168.56.115. **Limitation:** Photo of the screen.

C17 Check whether the PC is joined to the domain	
Where / who	**Windows 10, PowerShell** | Privilege: Any user
Purpose	Prove domain membership, which is required for domain sign-in.
Command	`systeminfo | findstr /B "Domain"`
Breaking it down	**Part** / **What it means** `systeminfo` — Prints a long report about the computer (takes 10–20 seconds). `|` — Pipe the report into the next command. `findstr` — Windows search tool (note the **d** in findstr; I first typed finstr and got an error). `/B "Domain"` — Show only lines that *begin* with the word Domain.
Expected result	Domain: HopeLabs.Local (joined) or Domain: WORKGROUP (not joined).
Observed in this lab	Domain: HopeLabs.Local after correcting the finstr typo (Figure 24).
How to verify	Read the value after Domain:.
If it fails	WORKGROUP → join the PC to the domain first (Chapter 15, Part A).

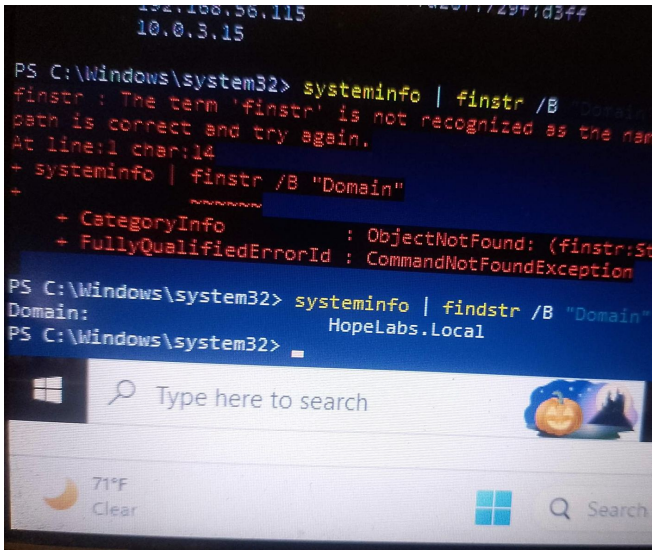


Figure 24 (Evidence E-24). Windows 10 is joined to HopeLabs.Local (C17)
Shows: The finstr typo error, then the corrected command returning Domain: HopeLabs.Local. **Limitation:** Photo of the screen.

6.9 Step 9 — Sign in as svc_backup on Windows 10

Where: Windows 10 sign-in screen. Choose **Other user**, type H0PELABS\svc_backup and the lab password, then sign out afterwards. **Expected:** the desktop loads. **Observed:** the lab owner reported a successful sign-in; the proof is the log evidence in the next step.

6.10 Step 10 — Prove the new account authenticated (event 4624)

Where: Wazuh dashboard. A domain user who signs in at a workstation is recorded **on the workstation**, so select the agent **DESKTOP-T6T1N2V** (006), not the Domain Controller.

Q4	`data.win.system.eventID: 4624 and data.win.eventdata.targetUserName: svc_backup`
Meaning	Event 4624 (successful logon) *and* the account that logged on is svc_backup. and requires both.
Used to	show that the rogue account was used.

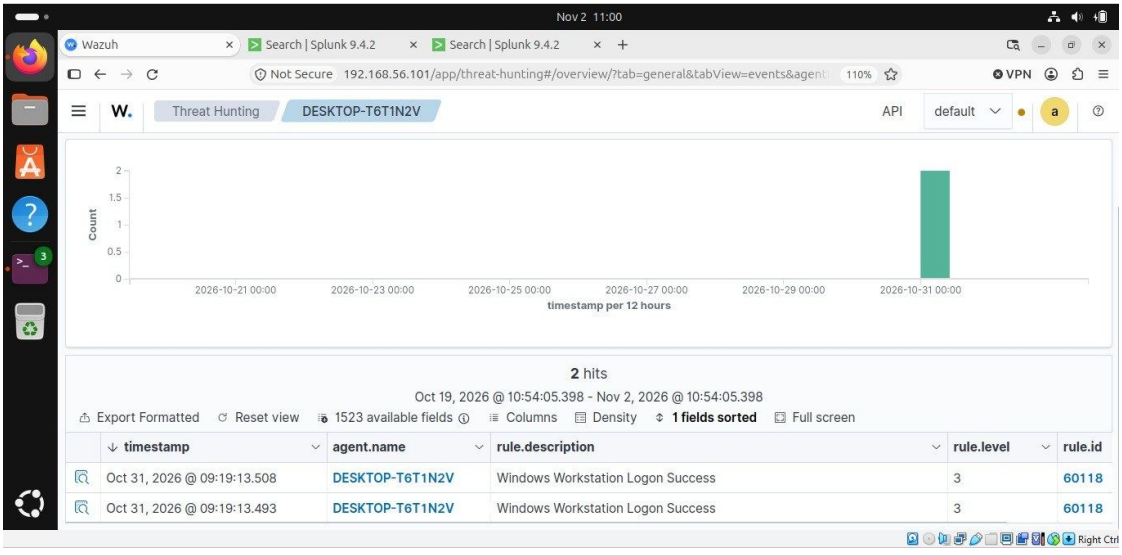


Figure 25 (Evidence E-25). Two Windows Workstation Logon Success alerts on 31 October
Shows: agent DESKTOP-T6T1N2V, rule 60118 "Windows Workstation Logon Success", level 3, 2 hits at 09:19:13. **Limitation:** Alert time (09:19) is later than the event time (09:06), see E-31.

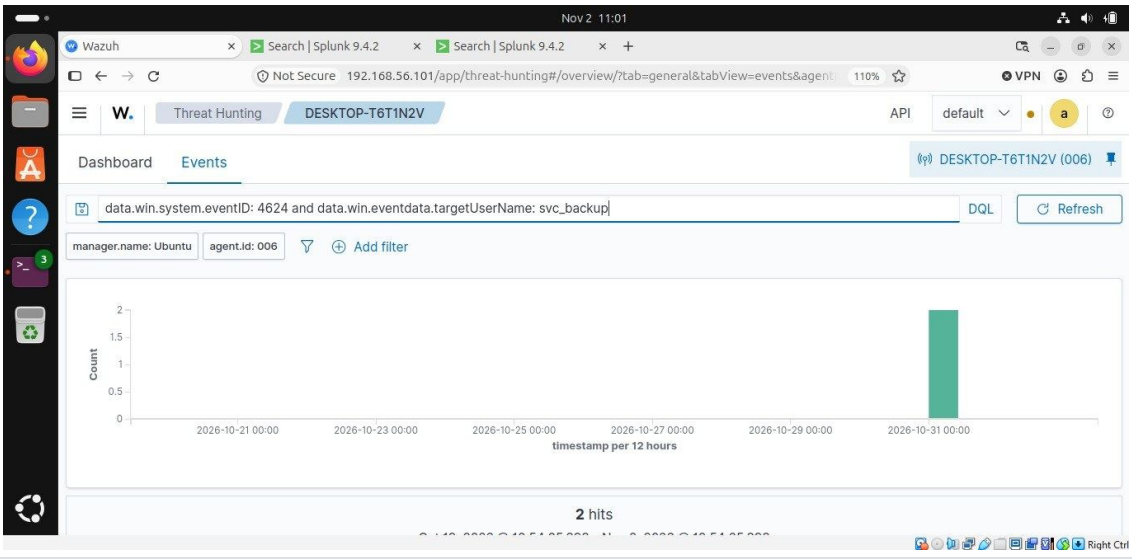


Figure 26 (Evidence E-26). The search that found them (Q4)
Shows: Agent 006, query for event 4624 and targetUserName svc_backup, 2 hits.

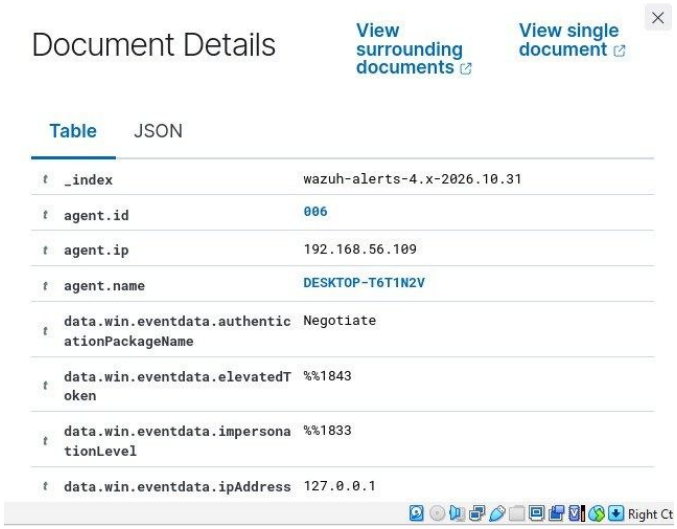


Figure 27 (Evidence E-27). Logon source
Shows: ipAddress 127.0.0.1 and authentication package Negotiate: the sign-in was local, not from another machine.

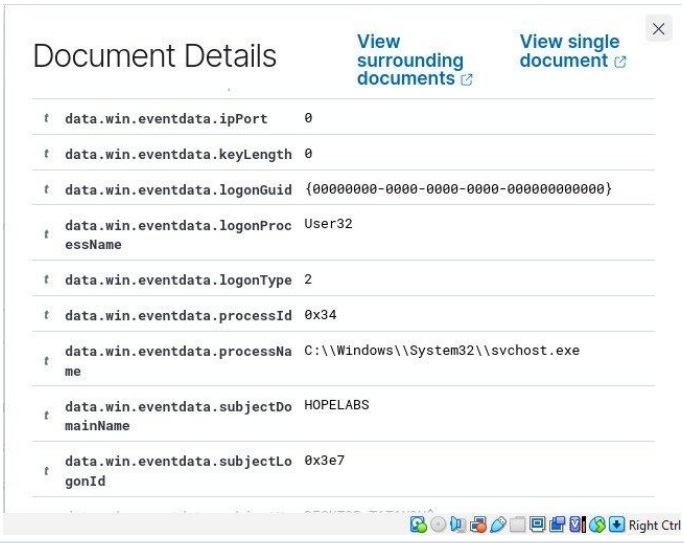


Figure 28 (Evidence E-28). Logon type
Shows: logonType 2 (interactive), logon process User32, process C:\Windows\System32\svchost.exe.

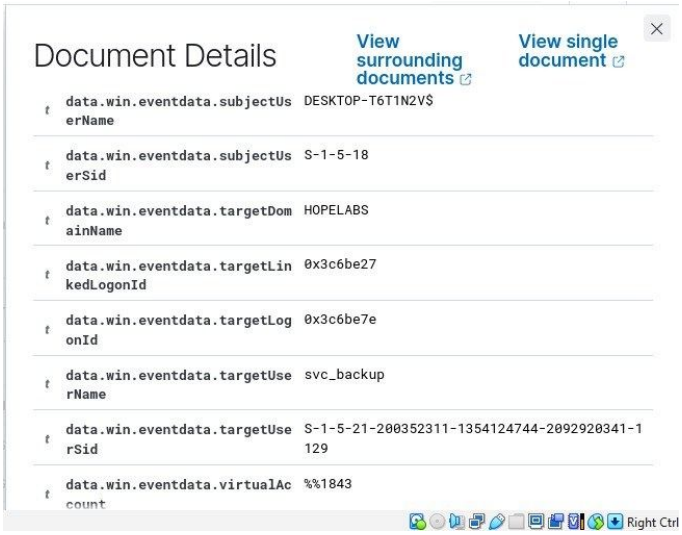


Figure 29 (Evidence E-29). Account and linked session

Shows: targetUserName svc_backup, domain HOPELABS, SID ending -1129; targetLogonId 0x3c6be7e linked to 0x3c6be27; subject DESKTOP-T6T1N2V$ (SYSTEM).

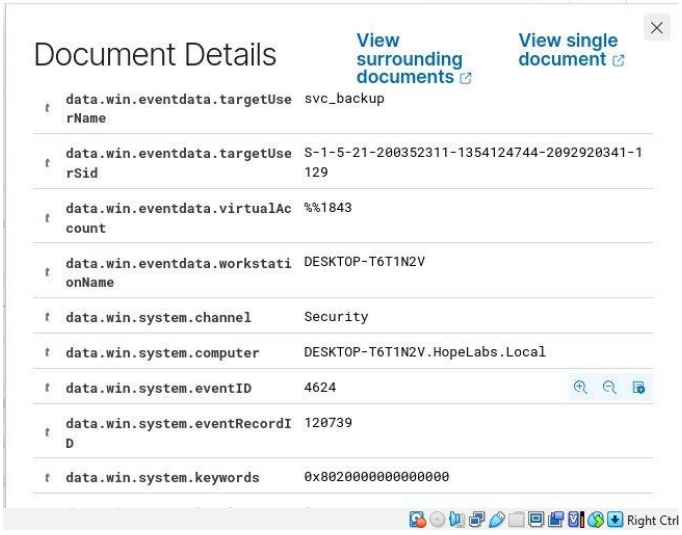


Figure 30 (Evidence E-30). Event identity

Shows: eventID 4624, workstationName DESKTOP-T6T1N2V, record 120739.

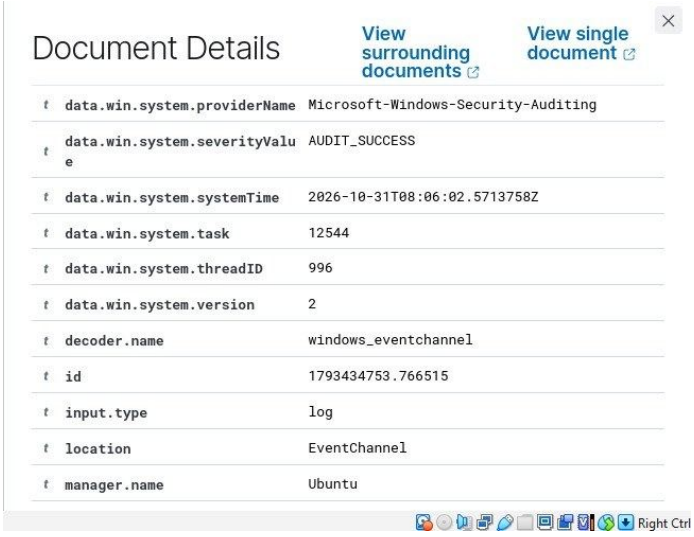


Figure 31 (Evidence E-31). The event's own time

Shows: systemTime 2026-10-31T08:06:02Z = 09:06:02 local.

Interpretation. A linked logon ID pair is consistent with a split-token (elevated and filtered) logon for an administrator account; this was not independently verified. The key facts: the account **did** authenticate, interactively, on DESKTOP-T6T1N2V, two days after promotion. The alert reached Wazuh about 13 minutes after the event; the reason (for example the VM not forwarding in real time) was not investigated.

> **EVIDENCE GAP.** Searching 4624 for svc_backup on the Domain Controller agent returned "No results" (Figure 40), but the Refresh button may not have been pressed before that photo, so the DC result is **inconclusive**, not a confirmed absence.

6.11 Step 11 — Re-investigate "blind" like a real analyst

A real analyst does not know which account is bad. So the investigation was repeated from the alerts only, starting with the workstation, then the Domain Controller.

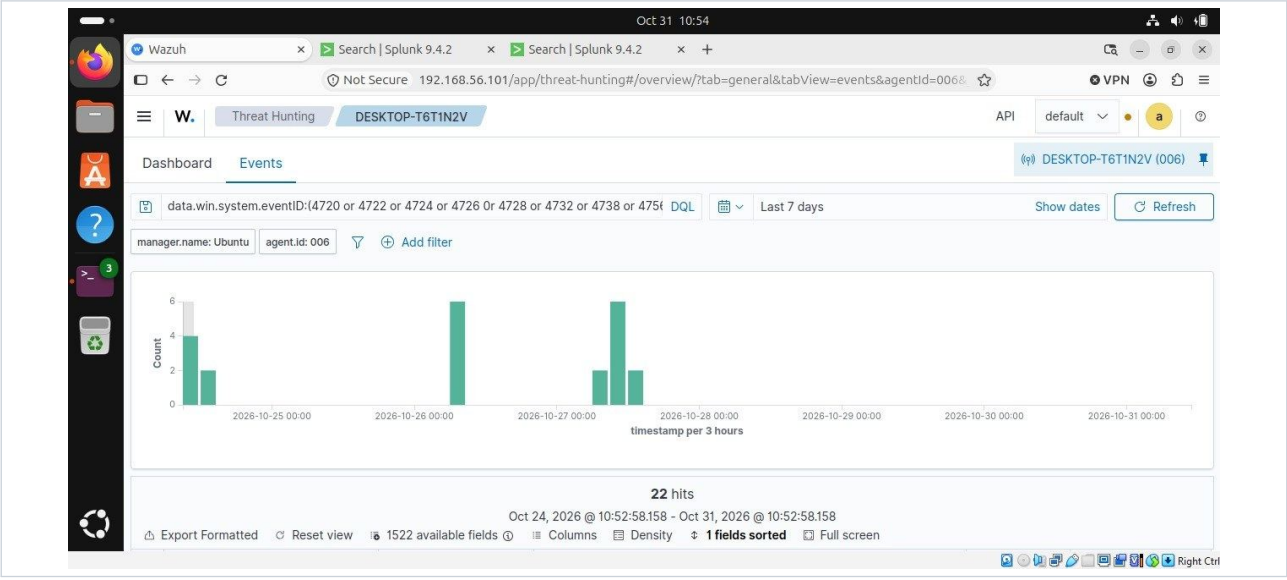


Figure 32 (Evidence E-32). Windows 10 identity changes over 7 days: 22 hits

Shows: Query for identity event IDs on agent 006 returned 22 hits spread across 24, 26 and 27 October — all before svc_backup existed. **Limitation:** The query contained a typo ("0r"), but the results still contained 4738 events.



Figure 33 (Evidence E-33). A baseline event: SYSTEM changing the local user 'aigbo'

Shows: Message of a 4738 event: logon ID 0x3E7 (SYSTEM), target account aigbo on DESKTOP-T6T1N2V. **Limitation:** Only one of the 22 events was opened; the others showed the same rule 60110 pattern.

Verdict: the workstation changes were SYSTEM adjusting a local account's display name — routine, not evidence of compromise. Lesson: always check *who* (subject) and *what changed* before escalating.

Q5	`data.win.eventdata.targetUserName: svc_backup and data.win.system.eventID: 4724`
Meaning	Password-reset events (4724) aimed at svc_backup.
Used to	check whether anyone reset the account's password after creation.

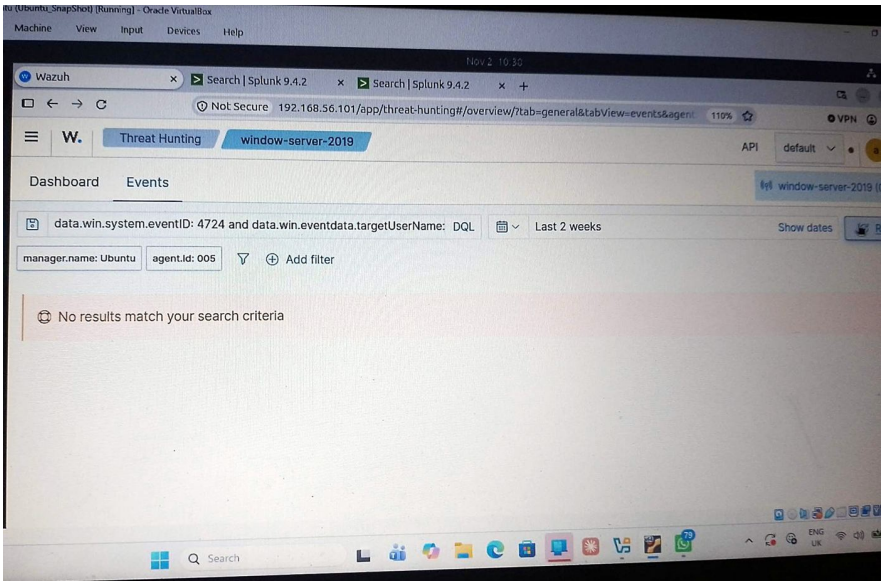


Figure 34 (Evidence E-34). No password reset found (4724)

Shows: Zero results for event 4724 on svc_backup over the last two weeks on the DC agent. **Limitation:** A negative result depends on the query and refresh being correct.

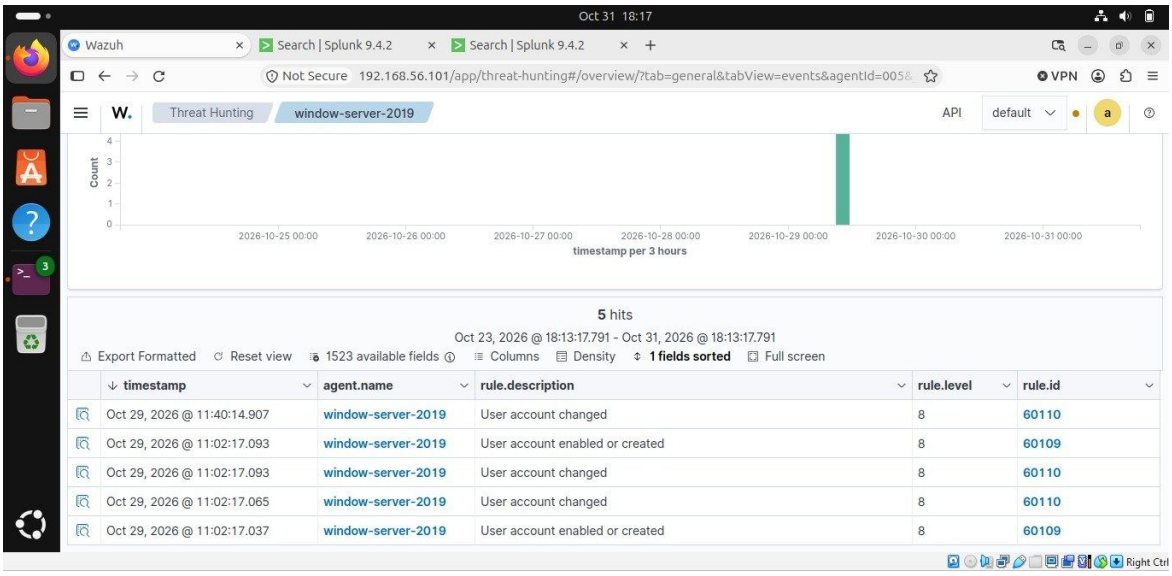


Figure 35 (Evidence E-35). Domain Controller identity alerts: 5 hits on 29 October

Shows: Two 60109 and three 60110 alerts. The newest row, 11:40:14.907 (60110), is unlike the others.

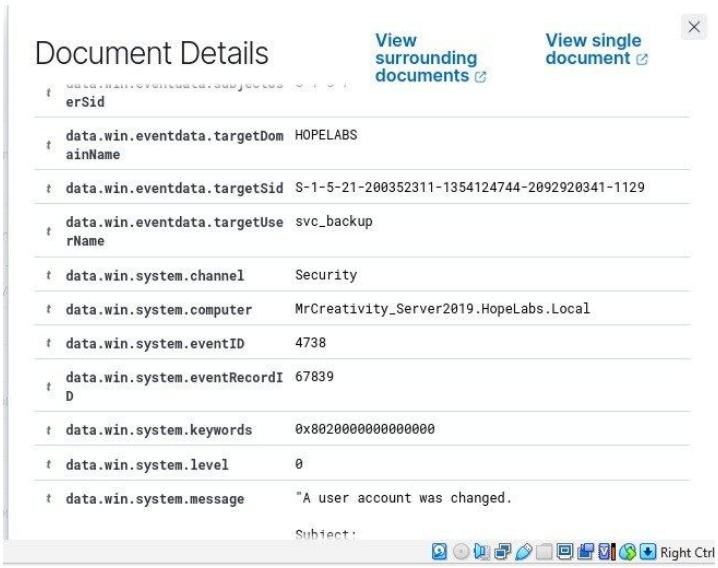


Figure 36 (Evidence E-36). Event 4738 on svc_backup, record 67839
Shows: targetUserName svc_backup, SID ending -1129, computer MrCreativity_Server2019.HopeLabs.Local, event 4738.



Figure 37 (Evidence E-37). Session of the 11:39 change
Shows: Message: Logon ID 0x3E6, target account svc_backup (SID ending -1129). **Limitation:** 0x3E6 is the logon ID of the built-in ANONYMOUS LOGON session.

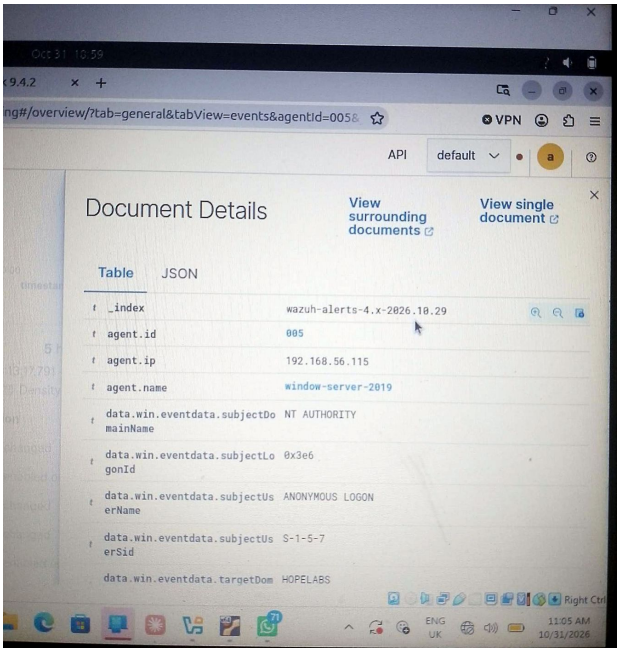


Figure 38 (Evidence E-38). The initiator was ANONYMOUS LOGON
Shows: subjectUserName ANONYMOUS LOGON, SID S-1-5-7, domain NT AUTHORITY, subject logon ID 0x3e6, agent 005.
Limitation: Photo of the screen.

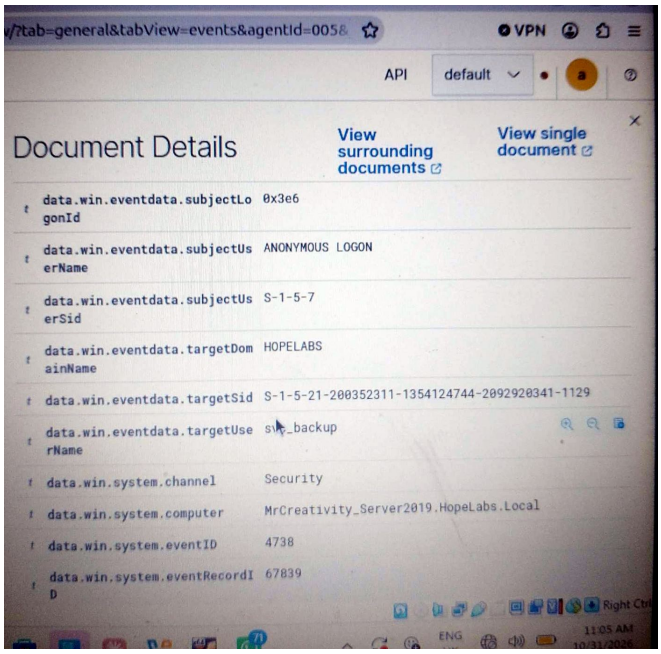


Figure 39 (Evidence E-39). The 11:39 event carries no changed attributes

Shows: Between agent.name and subjectDomainName the event contains no displayName, UAC or password fields; only the target (svc_backup) is present. **Limitation:** Absence of fields is the observation; why Windows logged it so was not determined.

Assessment: unexplained, low-to-medium. An anonymous session touching a brand-new, privileged account is worth a note; but the event holds no changed values, so it cannot be tied to an attacker action.

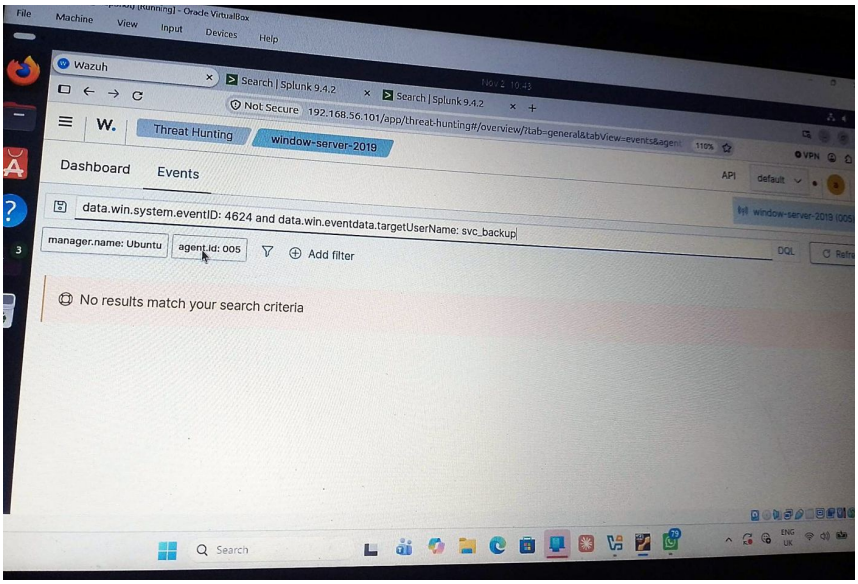


Figure 40 (Evidence E-40). Logon search for svc_backup on the DC (inconclusive)

Shows: Query for 4624 and svc_backup shown with "No results". **Limitation:** Refresh may not have been pressed; see the evidence-gap note in Step 10.

6.12 Step 12 — Review the alerts that followed the sign-in

Between 09:19 and 09:20 on 31 October the workstation produced further alerts (new service registry keys 92307, scripting files under user folders 92200, a Sysmon 'suspicious process' 61634 at level 12 and an 'abnormal command prompt' 92052 at level 4). They are triaged below by looking at the process, its parent and the command line.

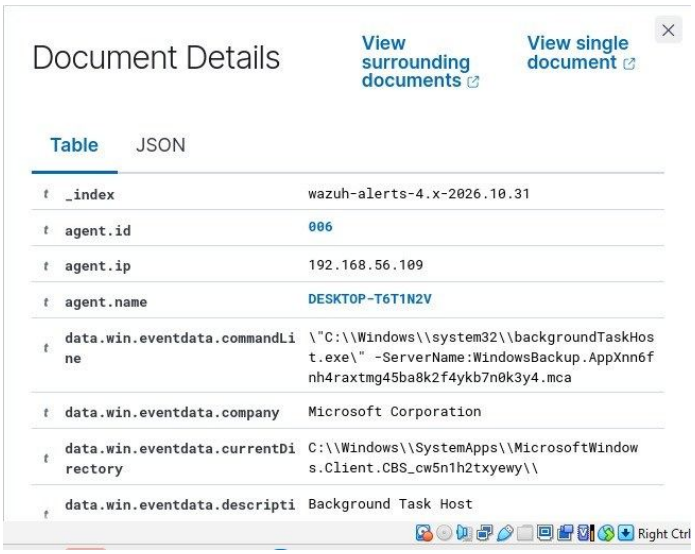


Figure 41 (Evidence E-41). Rule 61634 trigger: backgroundTaskHost.exe
Shows: commandLine "C:\Windows\system32\backgroundTaskHost.exe" -ServerName:WindowsBackup...; company Microsoft Corporation; directory under C:\Windows\SystemApps.

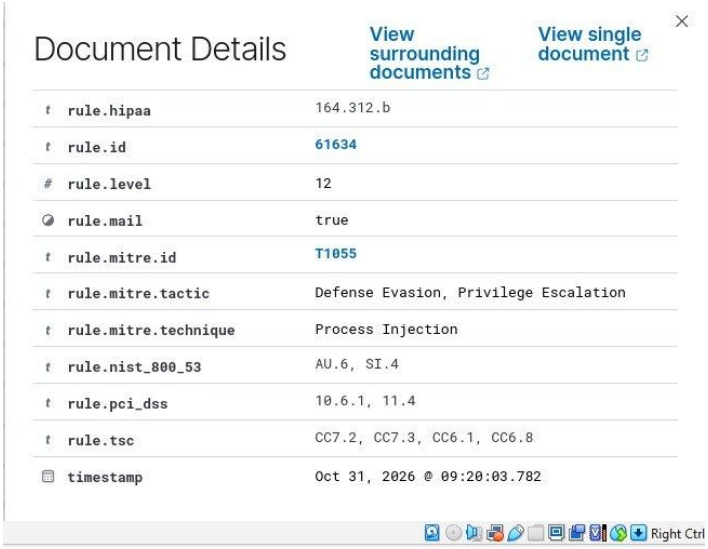


Figure 42 (Evidence E-42). Rule 61634: level 12, T1055 Process Injection
Shows: Rule 61634 level 12, mail true, MITRE T1055, timestamp 31 Oct 2026 09:20:03.782.

Verdict: false positive. The rule name mentions "taskhost.exe" and matched the text inside the legitimate Microsoft binary backgroundTaskHost.exe (correct path, Microsoft-signed metadata, medium integrity, normal parent). No injection evidence.

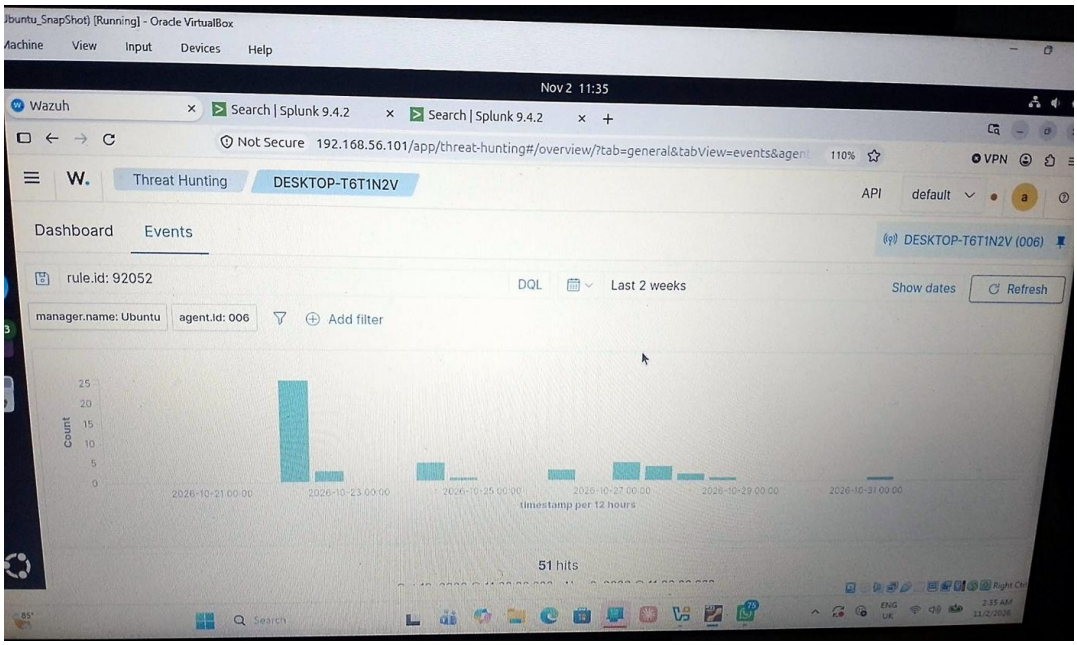


Figure 43 (Evidence E-43). Rule 92052 fires all the time (51 hits in two weeks)
Shows: The chart shows bars from 22 October onward and a very small bar on 31 October. **Limitation:** Photo of the screen.

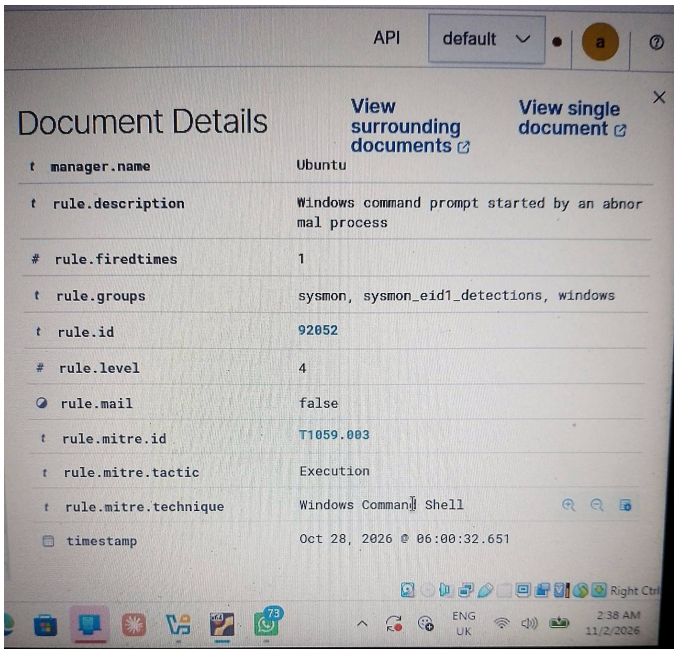


Figure 44 (Evidence E-44). A baseline 92052 event on 28 October (before svc_backup existed)
Shows: Rule 92052 level 4, T1059.003, timestamp 28 Oct 2026 06:00:32.

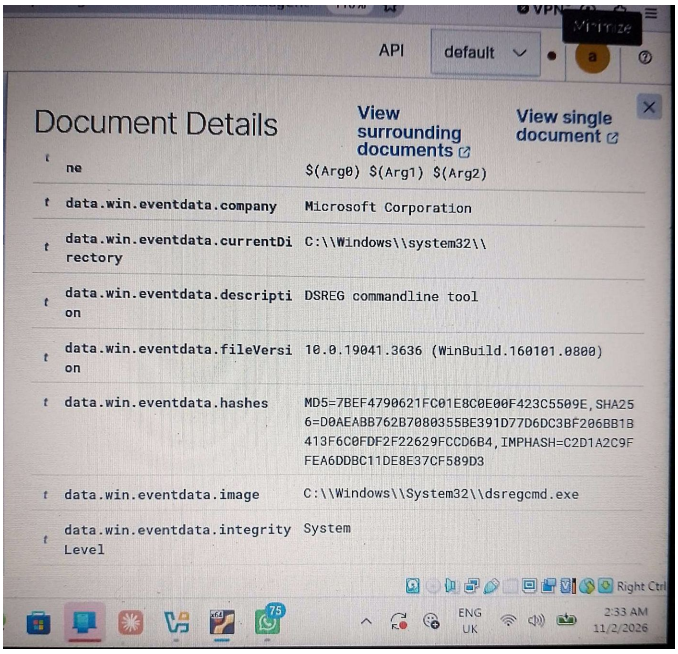


Figure 45 (Evidence E-45). The process: dsregcmd.exe

Shows: C:\Windows\System32\dsregcmd.exe, description "DSREG commandline tool", Microsoft Corporation, integrity System.

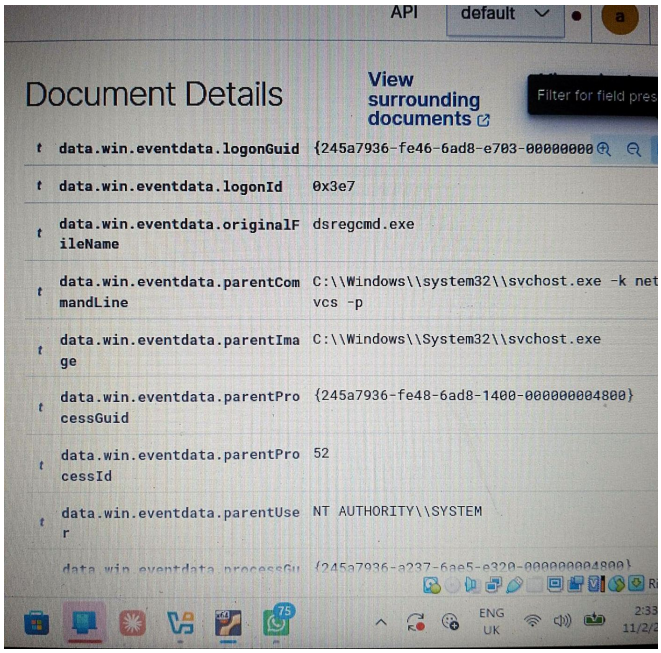


Figure 46 (Evidence E-46). The parent: svchost.exe as SYSTEM

Shows: parentImage C:\Windows\System32\svchost.exe, parentCommandLine svchost.exe -k netsvcs -p, parentUser NT AUTHORITY\SYSTEM, logon ID 0x3e7.

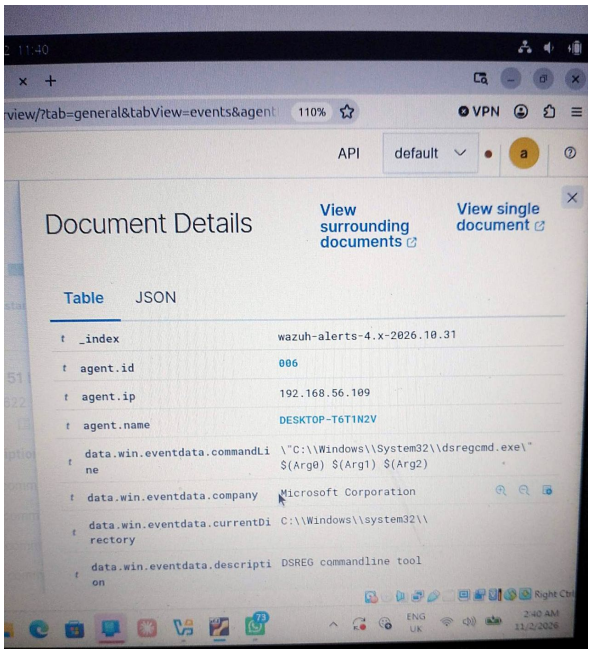


Figure 47 (Evidence E-47). The 31 October 92052 event (command line)

Shows: dsregcmd.exe $(Arg0) $(Arg1) $(Arg2) — the unfilled placeholders used by a scheduled task — index wazuh-alerts-2026.10.31.

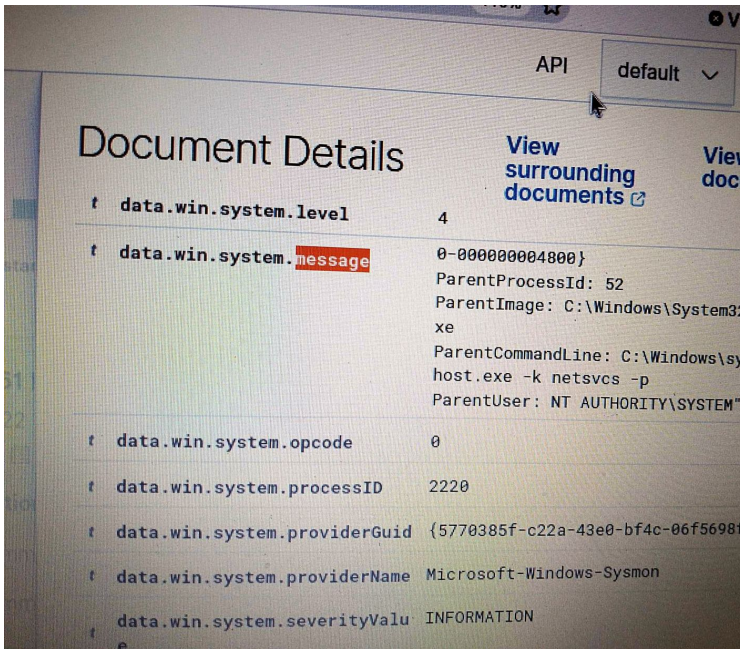


Figure 48 (Evidence E-48). The 31 October event's parent

Shows: Sysmon message: ParentImage svchost.exe, ParentCommandLine svchost.exe -k netsvcs -p, ParentUser NT AUTHORITY\SYSTEM (UtcTime 08:09:27).

Verdict: false positive. The 31 October event is identical in pattern to the baseline from 28 October: a Microsoft device-registration tool launched by the Task Scheduler service as SYSTEM. It is not connected to svc_backup (different session).

6.13 Step 13 — Write the custom detection rule (rule 100500)

Why: Wazuh raised the creation (level 8) and the promotion (level 12) as *separate* alerts. A SOC wants one high-priority alert when both happen close together. **Where:** the Ubuntu VM terminal (Wazuh manager). The rule itself is explained line by line in Chapter 7.

C18 Back up the rules file

Where / who	**Ubuntu VM, terminal** | Privilege: sudo (administrator)
Purpose	Keep a restore point before editing.
Command	`sudo cp /var/ossec/etc/rules/local_rules.xml \ /var/ossec/etc/rules/local_rules.xml.bak`
Breaking it down	

Part	What it means
`sudo`	Run the command as administrator; you will be asked for your password (nothing appears as you type).
`cp`	Copy a file.
`/var/ossec/etc/rules/local_rules.xml`	The file where you place your own Wazuh rules. Source.
`\ (backslash at line end)`	Continues the command on the next line.
`...local_rules.xml.bak`	Destination: same name with .bak added.

Expected result	No output.
Observed in this lab	The lab owner reported completing this step; no terminal screenshot was captured.
How to verify	Run `ls /var/ossec/etc/rules/` and look for local_rules.xml.bak.
If it fails	"No such file" → check the path; "Permission denied" → you forgot sudo.

C19 Open the rules file and add the rule

Where / who	**Ubuntu VM, terminal** | Privilege: sudo
Purpose	Add rule 100500 at the very end of the file.
Command	`sudo nano /var/ossec/etc/rules/local_rules.xml`
Breaking it down	

Part	What it means
`nano`	A simple text editor that runs in the terminal.
`Ctrl+O, Enter`	Save ("write Out") and confirm the file name.
`Ctrl+X`	Exit nano.
`Arrow keys / Ctrl+V`	Move around; Ctrl+V jumps down a page.

Expected result	The file opens. Scroll to the bottom and paste the rule from Chapter 7 (Listing 1).
Observed in this lab	Rule added; its effect is proved by Figures 51–53.
How to verify	Reopen the file and check the pasted text.
If it fails	Do not reuse an ID that already exists (100500 was free). Every opening tag needs a closing tag.

C20 Check the rules for syntax errors

Where / who	**Ubuntu VM, terminal** | Privilege: sudo
Purpose	Catch mistakes before restarting the manager.
Command	`sudo /var/ossec/bin/wazuh-analysisd -t`
Breaking it down	

Part	What it means
`/var/ossec/bin/wazuh-analysisd`	Wazuh's analysis engine, the program that loads and runs the rules.
`-t`	Test mode: load the configuration and rules, report errors, then exit.

Expected result	No ERROR lines (or a short success summary).
Observed in this lab	No output was captured; the rule fired later, which shows it loaded.

C20 Check the rules for syntax errors	
How to verify	Any ERROR names a file and line number.
If it fails	Fix the line mentioned, save, run again. Do not restart until it is clean.

C21 Restart the manager and confirm it is running	
Where / who	**Ubuntu VM, terminal** | Privilege: sudo
Purpose	Make Wazuh load the new rule.
Command	`sudo systemctl restart wazuh-manager` `sudo systemctl status wazuh-manager --no-pager | head -5`
Breaking it down	**Part** / **What it means** `systemctl` — Controls background services on Ubuntu. `restart wazuh-manager` — Stop and start the Wazuh manager service. `status` — Show the current state. `--no-pager` — Print all output at once instead of opening a scrolling viewer. `| head -5` — Show only the first five lines.
Expected result	The status block contains "active (running)".
Observed in this lab	No terminal screenshot was captured; the later alert proves the manager restarted with the rule.
How to verify	Look for active (running).
If it fails	"failed" → run C20 and fix the error; restore the backup with `sudo cp local_rules.xml.bak local_rules.xml` in the same folder.

EVIDENCE GAP. Steps C18–C21 were carried out and reported by the lab owner but not screenshotted. Evidence for the rule is indirect: alert 100500 reproduces the rule's description, level 13, groups and MITRE IDs (Figures 52 and 53).

6.14 Step 14 — Prove the rule with a second test account

Repeat the attack with a *different* name (svc_test2) so it is easy to tell from svc_backup. Run C4, C5, C6 (using svc_test2 and a different lab password), then C7 and C8 (with svc_test2) within 30 minutes of each other.

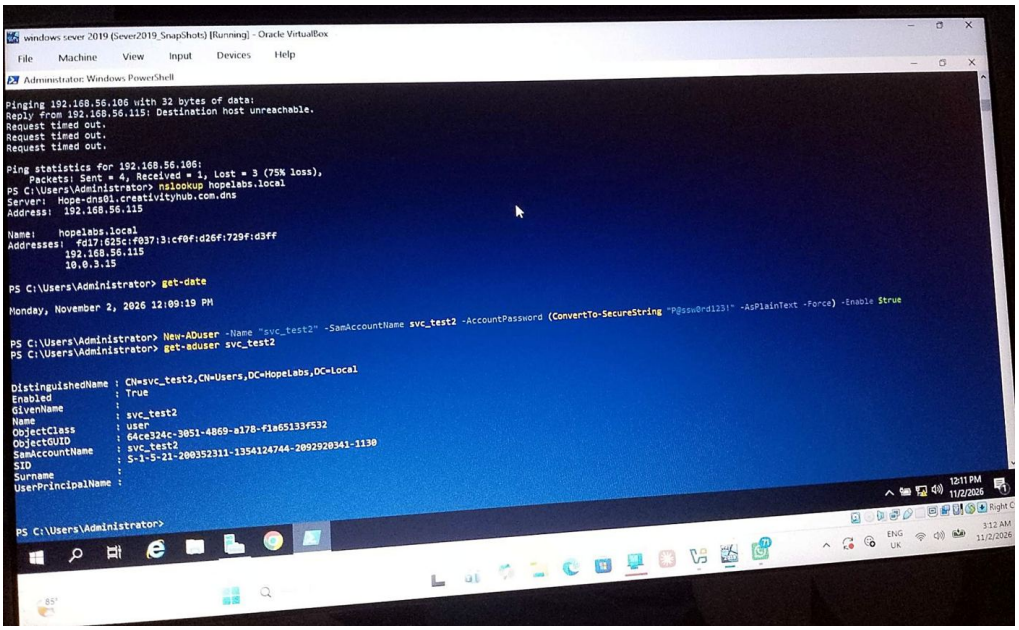


Figure 49 (Evidence E-49). svc_test2 created

Shows: Get-Date, New-ADUser svc_test2 and Get-ADUser: Enabled True, SID ending -1130. **Limitation:** Get-Date printed 12:09:19 PM but the taskbar read 12:11 PM when Get-ADUser ran; the account was created in between.

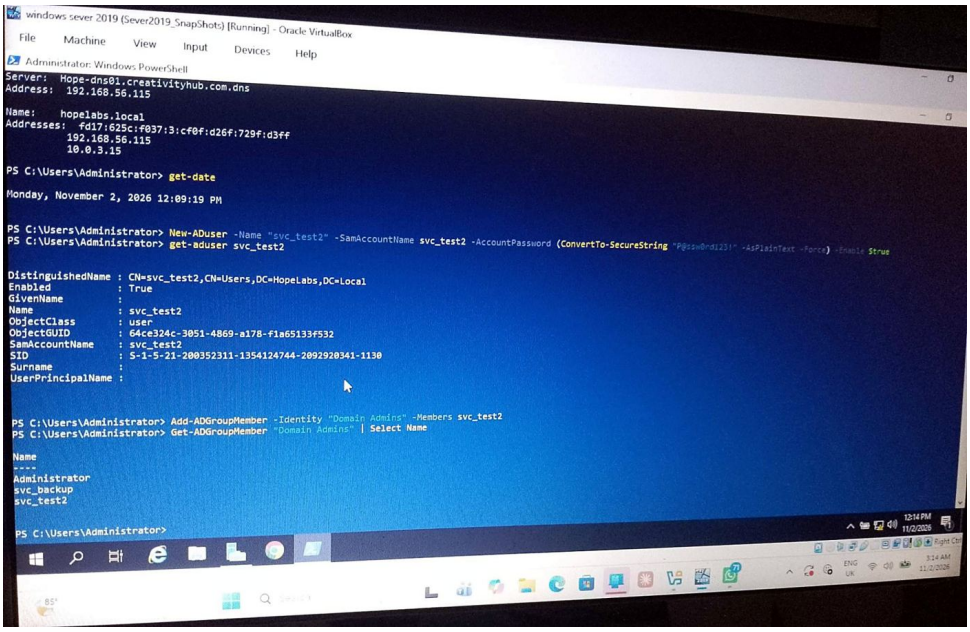


Figure 50 (Evidence E-50). svc_test2 added to Domain Admins

Shows: Add-ADGroupMember then Get-ADGroupMember listing Administrator, svc_backup and svc_test2 at 12:14 PM.

Q6	`rule.id: 100500`
Meaning	Show only alerts produced by the custom rule.
Used to	prove that the detection fires.

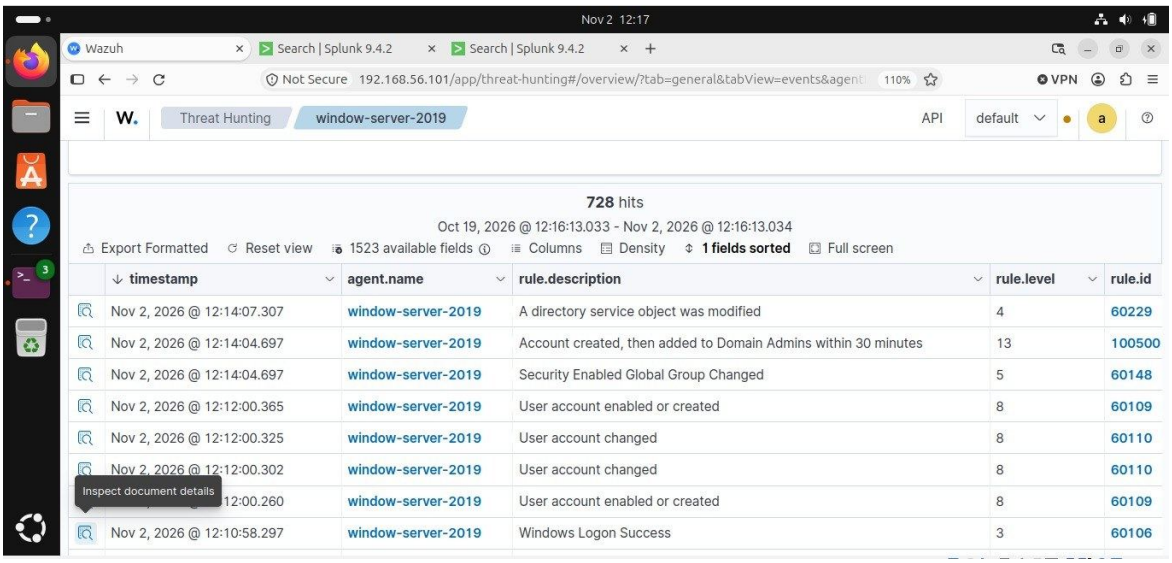


Figure 51 (Evidence E-51). Rule 100500 fired at 12:14:04

Shows: Row "Account created, then added to Domain Admins within 30 minutes", level 13, between 60148 (Security Enabled Global Group Changed, level 5) and the 60109/60110 account alerts. **Limitation:** A 60159 row for svc_test2 is not visible in this capture; why the default Domain Admins alert is not listed alongside 100500 was not investigated.

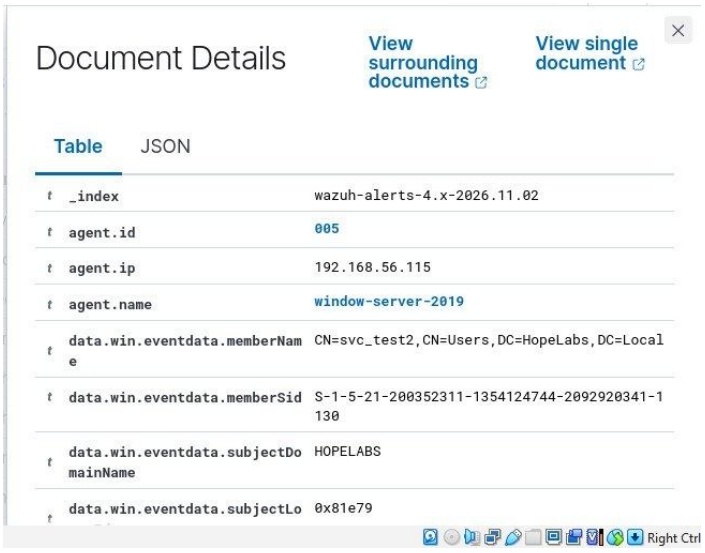


Figure 52 (Evidence E-52). The alert's data: svc_test2 added

Shows: memberName CN=svc_test2,CN=Users,DC=HopeLabs,DC=Local; member SID ending -1130; subject HOPELABS, logon ID 0x81e79; agent 005.

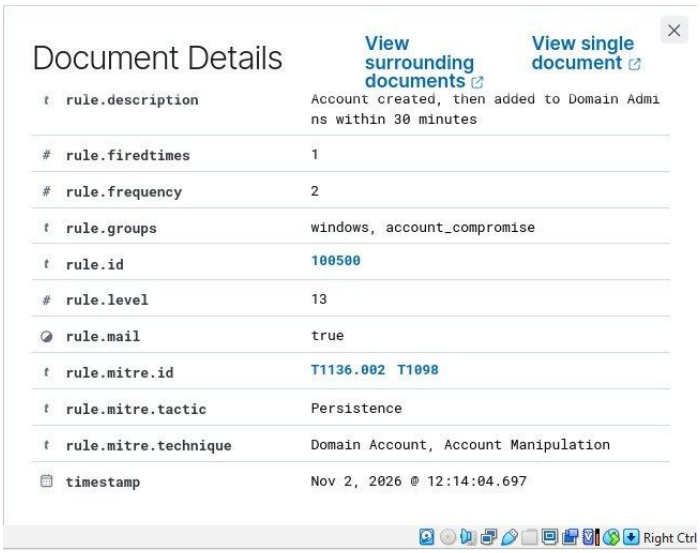


Figure 53 (Evidence E-53). Rule 100500 details

Shows: rule.id 100500, level 13, rule.frequency 2, groups windows and account_compromise, MITRE T1136.002 and T1098, tactic Persistence, timestamp 2 Nov 2026 12:14:04.697. **Limitation:** The test shows the rule fires; false-positive and negative tests were not run.

Chapter 7 — Configuration and Detection Logic

7.1 The custom rule (Listing 1)

The rule lives in `/var/ossec/etc/rules/local_rules.xml` on the Wazuh manager. It was added at the end of the file; nothing else in the file was changed.

```
<group name="windows,account_compromise,">
  <rule id="100500" level="13" timeframe="1800">
    <if_sid>60159</if_sid>
    <if_matched_sid>60109</if_matched_sid>
    <description>Account created, then added to Domain Admins within 30 minutes</description>
    <mitre>
      <id>T1136.002</id>
      <id>T1098</id>
    </mitre>
  </rule>
</group>
```

Listing 1 — rule 100500, exactly as entered.

Table 5. Rule 100500, line by line

Line	What it does
`<group name="windows,account_compromise,">`	A folder for rules. The names become *tags* on the alerts; the dashboard showed "windows, account_compromise" (Figure 53).
`<rule id="100500" level="13" timeframe="1800">`	id: unique number (Wazuh reserves 100000 and above for custom rules). level: severity 0–15; 13 is above anything else seen in the lab (12). timeframe: look back 1800 seconds = 30 minutes.
`<if_sid>60159</if_sid>`	The **trigger**: the rule is evaluated only when rule 60159 ("Domain Admins Group Changed") has just fired.
`<if_matched_sid>60109</if_matched_sid>`	The **memory**: rule 60109 ("User account enabled or created") must also have fired within the timeframe. This is what links creation and promotion.
`<description>`	Text shown in the dashboard alert.
`<mitre> <id>T1136.002</id> <id>T1098</id>`	Tags the alert with MITRE techniques Create Domain Account and Account Manipulation.
closing tags	Every tag opened must be closed; a missing one makes wazuh-analysisd -t report an error (C20).

7.2 Original configuration, actual change, recommended improvement

Table 6. Original, actual and recommended configuration

Category	Content
Original	Default Wazuh Windows rules: 60109/60110 (account created/changed, level 8), 60148 (security-enabled global group changed, level 5) and 60159 (Domain Admins group changed, level 12). They alert independently.
Actual change	One added rule (100500, level 13). Audit policy enabled on the DC (C1, C2). DNS and interface metrics changed on Windows 10 (C14, C16).
Recommended (not done)	Link by **account name**. The creation alert stores the name in `targetUserName` while the group alert stores it in `memberName`, so Wazuh's same_field option cannot match them directly; a custom decoder or a SIEM correlation search would be needed. Add an allow-list for approved administrators, and a second rule for any 4728 where the group is Domain Admins regardless of the creation. These ideas are untested.

7.3 Behaviour and limits of the rule

- **Observed:** fired at 12:14:04.697 on 2 Nov 2026 with rule.frequency 2, level 13, tags windows/account_compromise (Figure 53).

- **Not account-specific:** any new account followed by any Domain Admins change within 30 minutes triggers it (possible false positive if one admin creates an unrelated user and a colleague promotes someone else).
- **Possible false negatives:** promotion more than 30 minutes after creation, or an account created by a path that does not raise 60109.
- **Retroactive test:** the rule was not replayed against the 29 October svc_backup events; the 14-minute gap there is inside the window.

7.4 Query reference and key fields

Table 7. Wazuh dashboard searches used

ID	Query	Purpose
Q1	data.win.system.channel: Security	Prove Security events arrive
Q2	data.win.system.eventID: (4720 or 4722 or 4724 or 4726 or 4728 or 4732 or 4738 or 4756)	All identity changes
Q3	data.win.system.eventID: 4728	Domain Admins promotion
Q4	data.win.system.eventID: 4624 and data.win.eventdata.targetUserName: svc_backup	Did the account sign in
Q5	data.win.eventdata.targetUserName: svc_backup and data.win.system.eventID: 4724	Password reset check
Q6	rule.id: 100500	Custom rule fired

Table 8. Important alert fields

Field	Meaning
data.win.system.eventID	Windows event number (4728, 4624 …)
data.win.eventdata.subjectUserName / subjectLogonId	Who performed the action and their session
data.win.eventdata.targetUserName	Who the action was performed on (or who logged on)
data.win.eventdata.memberName / memberSid	The account added to a group (4728)
data.win.eventdata.logonType / ipAddress / workstationName	How and from where a logon happened
data.win.system.systemTime	Time Windows wrote the event, in UTC
rule.id / rule.level / rule.mitre.*	Which Wazuh rule fired, its severity and ATT&CK mapping

Chapter 8 — Testing and Results

8.1 Test register

Table 9. Test register

#	Test	Expected	Actual	Status
T1	Enable account auditing (C1–C3)	Subcategories at Success and Failure	As expected (E-03)	**Verified**
T2	DC events reach Wazuh (Q1)	Security events visible	82 hits (E-04, E-05)	**Verified**
T3	Creation detected	Alert for new account	Rules 60109/60110, level 8 (E-07, E-08)	**Verified**
T4	Promotion detected	Alert for Domain Admins change	4728, rule 60159, level 12 (E-10 to E-14)	**Verified**
T5	New account authenticates	A 4624 event	4624 type 2 on DESKTOP-T6T1N2V (E-25 to E-31); DC search inconclusive (E-40)	**Verified** (workstation)
T6	Password reset after creation	Unknown	No 4724 found (E-34)	**Negative result**
T7	Account deletion detected	4726 alert	Not performed	**Not tested**
T8	Splunk cross-check	Same events in Splunk	No Splunk output captured	**Not tested**
T9	Post-logon alerts triaged	Classify 61634 and 92052	Both false positives (E-41 to E-48)	**Verified**
T10	Custom rule 100500	Fires on create + promote	Fired at 12:14:04 for svc_test2 (E-49 to E-53)	**Verified**
T11	Rule false-positive / negative tests	No alert without promotion	Not performed	**Not tested**
T12	Domain sign-in works	svc_backup signs in	Failed first, then fixed (E-15 to E-24)	**Verified**

8.2 Reconstructed timeline

Local time = UTC+1 (the event's own systemTime is UTC; for example 10:15:48Z appears as 11:16:45 in the dashboard because of the one-hour offset and about a minute of ingestion delay).

Table 10. Reconstructed timeline of identity activity

Local time	Event	Detail	Evidence
29 Oct 11:01:44	Account enabled (4722)	Administrator → svc_backup, alerts 60109/60110 at 11:02:17	E-07, E-08
29 Oct 11:15:48	Added to Domain Admins (4728)	Administrator (session 0x81e79) → svc_backup; rule 60159 level 12 at 11:16:45	E-10 to E-14
29 Oct 11:39:17	Account changed (4738)	ANONYMOUS LOGON (0x3E6); no attributes; alert 11:40:14	E-35 to E-39
31 Oct 09:06:02	Interactive logon (4624, type 2)	svc_backup on DESKTOP-T6T1N2V; alert 09:19:13 (13 min later)	E-25 to E-31
31 Oct 09:09:27	dsregcmd.exe (Sysmon 1)	SYSTEM scheduled task; alert 92052 (benign)	E-47, E-48
31 Oct 09:20:03	Alert 61634 level 12	backgroundTaskHost.exe (false positive)	E-41, E-42
2 Nov ~12:11	svc_test2 created	Test of the detection; alerts 60109/60110 at 12:12:00	E-49
2 Nov 12:14:04	svc_test2 promoted; rule 100500	Level 13 alert	E-50 to E-53

Chapter 9 — Evidence and Screenshot Analysis

9.1 Evidence integrity

- **Originals preserved.** The unmodified images remain in the uploaded evidence set and should be committed unchanged to the SOC-Home-Lab-Evidence repository.
- **Presentation copies.** Direct screenshots are included as-is (resized only if very large). Camera photos of a screen were automatically cropped to the screen area and given contrast/sharpness enhancement. No pixels were redrawn, no values altered, no annotations added, and failed commands/errors were left visible.
- **Sensitive data.** Lab-only passwords (for example in New-ADUser commands) are visible in some captures and were not redacted, to avoid altering evidence. They belong to disposable lab accounts; change or delete them before publishing publicly.
- **Duplicates.** Near-identical captures were omitted. Events were identified by reading the images; where a value could not be read reliably it is not used.

9.2 Evidence register

All figures appear in the chapter where they are used; this register lists them for lookup. Evidence IDs equal figure numbers.

Table 11. Evidence register

E-ID	Source file	Title	Limitation
E-01	`1000485304`	The lab brief that defines the objective, tasks, deliverables and analyst questions	It is the course brief, not lab output; its layout was used as the design reference for this report.
E-02	`generated diagram`	Laboratory architecture and data flow	Diagram drawn from documented configuration; not a screenshot.
E-03	`1000485451`	Audit policy enabled on the Domain Controller (C1–C3)	Taken from the PowerShell console; it proves the policy at that moment, not that events were later collected.
E-04	`1000486325`	Domain Controller Security-channel events in Wazuh (Q1)	Read at the time of the first check; the count grew afterwards.
E-05	`1000486326`	Sample of the Security alerts: Windows Logon Success (60106) and DC Logon Failure (60131)	These are routine logons, not the account activity yet.
E-06	`1000486330`	Creation and verification of svc_backup (C4–C6)	The screenshot shows the commands, not the Wazuh result; that comes next.
E-07	`1000486350`	First alerts for svc_backup at about 11:02 on 29 October	The list does not show the account name; open a row for that (next figure).

E-ID	Source file	Title	Limitation
E-08	1000486351	Details of the account-change alert (event 4738)	The exact meaning of the 0x11 → 0x10 flag change was not decoded; it coincides with the 4722 "account was enabled" event and is treated as part of enabling the account.
E-09	1000487033	svc_backup added to Domain Admins (C4, C7, C8)	—
E-10	1000487064	Wazuh alert for the group change — "Domain Admins Group Changed", level 12	Index time 11:16:45 is about a minute later than the event itself (E-14).
E-11	1000487075	Rule 60159 details	The rule's MITRE mapping (T1484) differs from the T1098 mapping of the account rules; both are Wazuh's own labels.
E-12	1000487066	Who was added, and by whom	—
E-13	1000487068	Which group changed	—
E-14	1000487074	The event's own timestamp	Use this time, not the dashboard time, for the timeline.
E-15	20261030_093014871832062862026863 5	Sign-in failure on Windows 10	A photo of the screen, auto-cropped and contrast-enhanced.
E-16	20261031_060956603506509479716629 5	DNS lookups failing on Windows 10	Photo of the screen; text is small but legible at full zoom.
E-17	20261031_061327629843142208630007 3	Adapters and addresses on Windows 10 (C12, C13)	Photo of the screen.
E-18	20261031_061831775959163442897554	Setting the DNS server and flushing the cache (C14, C15)	After this change the lookups still reported 10.0.2.3, which led to the metric fix. Photo of the screen.
E-19	20261031_062133381731181520216464 4	Domain Controller network configuration (C9 on the DC)	The same screen also shows an unrelated ping to 192.168.56.106 (75% loss); the host at that address was not identified.
E-20	20261031_062153288193550373236124 4	DNS on the Domain Controller itself	The DC registers its NAT address as well, a multi-homed side effect noted in Chapter 11.

E-ID	Source file	Title	Limitation
E-21	20261031_06232965590130839863366 11	Windows 10 asking the DC directly	Proves the DC's DNS works and that the workstation was simply asking the wrong server.
E-22	20261031_06395540060366607975670 03	Typing errors while setting the metric	Errors left in on purpose; they are real troubleshooting evidence. Photo of the screen.
E-23	20261031_06432136790123557521366 13	DNS now answered by the Domain Controller	Photo of the screen.
E-24	20261031_06460842106901678780139 21	Windows 10 is joined to HopeLabs.Local (C17)	Photo of the screen.
E-25	1000490910	Two Windows Workstation Logon Success alerts on 31 October	Alert time (09:19) is later than the event time (09:06), see E-31.
E-26	1000490911	The search that found them (Q4)	—
E-27	1000490912	Logon source	—
E-28	1000490913	Logon type	—
E-29	1000490914	Account and linked session	—
E-30	1000490915	Event identity	—
E-31	1000490916	The event's own time	—
E-32	1000488783	Windows 10 identity changes over 7 days: 22 hits	The query contained a typo ("0r"), but the results still contained 4738 events.
E-33	1000488791	A baseline event: SYSTEM changing the local user 'aigbo'	Only one of the 22 events was opened; the others showed the same rule 60110 pattern.
E-34	20261102_10385049621339005234048 14	No password reset found (4724)	A negative result depends on the query and refresh being correct.
E-35	1000489405	Domain Controller identity alerts: 5 hits on 29 October	—
E-36	1000489407	Event 4738 on svc_backup, record 67839	—
E-37	1000489444	Session of the 11:39 change	0x3E6 is the logon ID of the built-in ANONYMOUS LOGON session.
E-38	20261031_19050470194579359863466 69	The initiator was ANONYMOUS LOGON	Photo of the screen.
E-39	20261031_19052512595383301389443 99	The 11:39 event carries no changed attributes	Absence of fields is the observation; why Windows logged it so was not determined.
E-40	20261102_10434652830735765871350 34	Logon search for svc_backup on the DC (inconclusive)	Refresh may not have been pressed; see the evidence-gap note in Step 10.
E-41	1000490945	Rule 61634 trigger: backgroundTaskHost.exe	—

E-ID	Source file	Title	Limitation
E-42	1000490951	Rule 61634: level 12, T1055 Process Injection	—
E-43	20261102_113545867362376990445656l	Rule 92052 fires all the time (51 hits in two weeks)	Photo of the screen.
E-44	20261102_113810230430018098292257 6	A baseline 92052 event on 28 October (before svc_backup existed)	—
E-45	20261102_113314792612479715885249 0	The process: dsregcmd.exe	—
E-46	20261102_113332702282650369640643 5	The parent: svchost.exe as SYSTEM	—
E-47	20261102_114036102976034741895391 5	The 31 October 92052 event (command line)	—
E-48	20261102_114128518500682427795925 9	The 31 October event's parent	—
E-49	20261102_121226289077794660376886 8	svc_test2 created	Get-Date printed 12:09:19 PM but the taskbar read 12:11 PM when Get-ADUser ran; the account was created in between.
E-50	20261102_121448356341462308990197 0	svc_test2 added to Domain Admins	—
E-51	1000491004	Rule 100500 fired at 12:14:04	A 60159 row for svc_test2 is not visible in this capture; why the default Domain Admins alert is not listed alongside 100500 was not investigated.
E-52	1000491005	The alert's data: svc_test2 added	—
E-53	1000491007	Rule 100500 details	The test shows the rule fires; false-positive and negative tests were not run.

Chapter 10 — Troubleshooting and Root-Cause Analysis

Table 12. Problems, causes and fixes

Problem	Symptom	Cause	Fix	Status
Wrong category name	auditpol: Error 0x00000057 (E-03)	"User Account Management" is a subcategory; /category needs "Account Management"	Use the right name or auditpol /list	Resolved
Misspelt commands	get--aduser, finstr, set-netinterface, set-neiptinterface: CommandNotFound	Typing errors	Copy exact commands (C6, C16, C17)	Resolved
Missing hyphen	"A positional parameter cannot be found that accepts argument ..." (E-18, E-22)	Parameter written without leading hyphen	Add the hyphen	Resolved
Domain sign-in refused	"your domain isn't available" (E-15)	Windows 10 asked the NAT adapter's DNS (10.0.2.3) first; the host-only adapter had only placeholder DNS	C14, C15, C16	Resolved
Wrong agent searched	Searches on agent 006 returned workstation noise instead of DC events	Agent filter left on the previous agent	Switch agent before searching	Resolved
Query typo	"0r" (zero) instead of "or"	Typing	Retype with the letter o	Resolved
Stale "No results"	4624 search on DC empty (E-40)	Refresh may not have been pressed	Press Refresh; search the workstation agent	Inconclusive
Late alerts	Alert about 1 min after event (DC) and about 13 min (workstation)	Ingestion delay; workstation cause not investigated	Use systemTime for timelines	Open

10.1 Root-cause analysis: the domain sign-in failure

1. **Observation 1:** sign-in message says the domain is unavailable (E-15).
2. **Observation 2:** nslookup hopelabs.local returns "UnKnown 10.0.2.3 ... Non-existent domain" (E-16). 10.0.2.3 is the NAT adapter's DNS server, which does not know the lab domain.
3. **Observation 3:** the host-only adapter listed placeholder DNS entries (fec0:0:0:ffff::1–3) instead of the DC (E-16).
4. **Test:** asking the DC directly (nslookup hopelabs.local 192.168.56.115) works (E-21). So the DC's DNS is healthy and reachable.
5. **Fix 1:** set Ethernet 2's DNS to 192.168.56.115 (C14). Lookups still went to 10.0.2.3 (E-18).
6. **Fix 2:** lower the metric of Ethernet 2 (5) and raise it on Ethernet (100) (C16). nslookup then returned 192.168.56.115 (E-23) and the join check and sign-in followed (E-24).
7. **Conclusion (confirmed by the fix):** Windows queried the NAT adapter's DNS first because of adapter priority. The cause of the placeholder DNS values themselves was not investigated.

> **NOTE.** The DC also registers its NAT address 10.0.3.15 under hopelabs.local (E-20, E-21). A workstation that picks that address cannot reach the DC. It did not break the lab but is a standing risk (Chapter 11, F5).

Chapter 11 — Security Findings and Risk Analysis

Table 13. Security findings

ID	Finding	Evidence	Confidence	Impact / rating
F1	A new account was added to Domain Admins about 14 minutes after creation by the same administrator session	E-07 to E-14	Confirmed	**High**: full domain control; this is the classic persistence-plus-escalation pattern
F2	The new Domain Admin signed in interactively on a workstation	E-25 to E-31	Confirmed	**Medium–High**: admin credentials cached on a workstation increase theft risk
F3	Unexplained 4738 under ANONYMOUS LOGON with no changed attributes	E-35 to E-39	Observed, cause unknown	**Low–Medium**
F4	Rules 61634 and 92052 raised alerts for legitimate Windows binaries	E-41 to E-48	Confirmed false positives	**Low** (analyst time)
F5	The DC registers its NAT IP (10.0.3.15) in DNS	E-20, E-21	Observed	**Low** (lab design)
F6	Default rules did not correlate creation and promotion	E-07, E-10	Confirmed	**Medium** (detection gap, now partly closed by rule 100500)
F7	Alert latency up to 13 minutes on the workstation	E-25, E-31	Observed	**Low–Medium**

11.1 The analyst questions (from the lab brief)

Table 14. Analyst questions and answers

Question	Answer from the evidence
What happened?	svc_backup was created and enabled on 29 Oct, added to Domain Admins 14 minutes later, and used to sign in on a workstation on 31 Oct.
How was it detected?	Wazuh rules 60109/60110 (account), 60159 (Domain Admins, level 12), 60118 (logon); custom rule 100500 now correlates the first two.
What evidence supports the finding?	Events 4722, 4738, 4728, 4624; fields subject/target/memberName, logon IDs and systemTime (E-07 to E-31).
Which host and account were involved?	Host: Domain Controller MrCreativity_Server2019 (agent 005) and workstation DESKTOP-T6T1N2V (agent 006). Accounts: initiator HOPELABS\Administrator; new account svc_backup (SID ending -1129).
What other systems or events should be checked?	All 4720/4722 events on the DC for other new accounts; Administrator's logon (session 0x81e79 source); 4768/4769/4776 for svc_backup elsewhere; scheduled-task/service creation (4698/7045); logons of svc_backup on other machines.
What action would you take in a real SOC?	Treat as a potential incident: confirm with change management, disable svc_backup, remove it from Domain Admins, review the initiator's session and credentials, hunt for other new accounts, preserve logs, escalate per the incident plan.

11.2 Evidence still needed before declaring compromise

- Approved change request or admin confirmation for svc_backup.
- Source and authentication of the Administrator session 0x81e79 (4624/4648 on the DC).
- Who was at the Windows 10 console on 31 October.
- Process and command history on the DC (Sysmon was confirmed on Windows 10 only).
- Any activity by svc_backup beyond the single interactive logon.
- An explanation of the ANONYMOUS LOGON 4738 event.

> **NOTE.** Because the lab author performed the actions, the activity is known to be a training exercise. The assessment above is what a SOC analyst could conclude from logs alone.

Chapter 12 — Mitigation and Remediation

Table 15. Mitigations (implemented vs recommended)

Recommendation	Status	How to implement	Verification	Residual risk
Enable account and group auditing	**Implemented** (C1, C2)	auditpol commands on the DC	auditpol /get (E-03); events in Wazuh	Low
Correlation rule creation + promotion	**Implemented** (100500)	Listing 1	Alert fired for svc_test2 (E-51 to E-53)	Not account-specific (Section 7.3)
Fix DNS priority on dual-homed machines	**Implemented** (C14, C16)	DNS + interface metrics	nslookup answered by DC (E-23)	Settings may reset after changes to adapters
Remove test accounts	Recommended	Chapter 15, Part H cleanup commands (not executed in the evidence)	Get-ADGroupMember "Domain Admins" shows only intended members	None after removal
Restrict Domain Admins; use separate admin accounts and no admin logon on workstations	Recommended	Review membership; use tiered administration	Quarterly membership review; alert on new members	Insider or stolen-credential risk
Tune noisy rules 61634 and 92052	Recommended	Add exclusions for Microsoft-signed backgroundTaskHost.exe and the dsregcmd.exe scheduled task	Alert volume drops; true tests still alert	A real attack could mimic an excluded binary
Add allow-list and account-specific correlation	Recommended	Custom decoder or SIEM search (Section 7.2)	Replay with a legitimate and a malicious scenario	Maintenance of the allow-list
Stop the DC registering its NAT address	Recommended	Disable registration on the NAT adapter in DNS settings	nslookup shows one DC address	Low
Install Sysmon on the Domain Controller	Recommended	Deploy Sysmon with a tested configuration	Sysmon events from agent 005 appear	Added log volume

Chapter 13 — What I Learned

- **Concept:** I learned that detection is a chain of six stages and that a break anywhere (audit policy, agent, rule, search, refresh button) looks like "no alert".
- **Practical skills:** I enabled auditing with auditpol, created and promoted an AD account with PowerShell, used Wazuh's search to find events, and wrote and tested a correlation rule.
- **Evidence analysis:** I learned to read subject versus target, logon IDs, event time versus alert time, and to check who an action came from before escalating.
- **Troubleshooting:** I solved a domain sign-in failure by proving the DC's DNS worked, then fixing which adapter Windows asked first. Almost all of my other errors were typing mistakes, so I now copy commands and check hyphens.
- **Security reasoning:** two low or medium alerts that look harmless alone (creation level 8, promotion level 12) can be one incident; also, a level 12 alert can be a false positive (61634). Severity numbers are a starting point, not a verdict.
- **Honesty in reporting:** I learned to record inconclusive results (the DC 4624 search) and untested items rather than claim success.
- **Where I need to grow:** Splunk correlation, writing account-specific rules and decoders, Sysmon on the DC, and testing rules for false positives.

Chapter 14 — Practical Knowledge and Competencies

Table 16. Skills mapped to evidence

Skill	Task in this lab	Evidence	Level
Windows audit configuration	auditpol (C1–C3)	E-03	Demonstrated
Active Directory administration	New-ADUser, Add-ADGroupMember, Get-ADGroupMember	E-06, E-09	Demonstrated
SIEM searching and triage	Q1–Q6, alert review	E-04 to E-53	Demonstrated
Windows event interpretation	4722, 4728, 4738, 4624	E-08, E-12, E-30	Demonstrated
Detection engineering	Rule 100500	E-51 to E-53	Demonstrated (basic)
Network and DNS troubleshooting	C9–C17	E-15 to E-24	Demonstrated
Sysmon alert triage	61634, 92052	E-41 to E-48	Demonstrated
Splunk searching	—	none	Not demonstrated
Attack simulation from Kali	—	none	Not demonstrated

Recommended next exercises: repeat with a deletion (event 4726); replay the rule against both accounts; build a Splunk search for the same sequence; write an account-specific rule; add Sysmon to the DC.

Chapter 15 — Reproduction Guide (for a complete beginner)

This chapter is a start-to-finish runbook. Every command is explained in Chapter 6 under the command-card number shown (C1, C2, …); you can copy the commands from here and read the card if something is unclear. Do this only in your own isolated lab.

Part A — Initial state checklist (before you start)

- VirtualBox with three VMs on a **host-only** network 192.168.56.0/24 (plus an optional NAT adapter each).
- Windows Server 2019: Domain Controller for **HopeLabs.Local**, static IP 192.168.56.115, DNS server role running.
- Windows 10: joined to HopeLabs.Local. If it is not joined (C17 shows WORKGROUP), join it:

```
Add-Computer -DomainName HopeLabs.Local -Credential HOPELABS\Administrator -Restart
```

Add-Computer joins this PC to a domain; **-DomainName** is the domain; **-Credential** prompts for the domain administrator's password; **-Restart** reboots to finish. This standard cmdlet was *not* run in this lab (the PC was already joined). It requires working DNS (C14, C16).

- Ubuntu: Wazuh manager and dashboard running; Wazuh agents installed and **Active** on both Windows machines.
- **Take a snapshot of each VM.** Write down: DC IP, workstation IP, Wazuh dashboard address.

Part B to G — Runbook

Table 17. Runbook

#	Where	Command / action	Card	Expected
1	DC	auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable	C1	"successfully executed"
2	DC	auditpol /set /subcategory:"Security Group Management" /success:enable /failure:enable	C2	"successfully executed"
3	DC	auditpol /get /category:"Account Management"	C3	Success and Failure on all six
4	Wazuh	Threat Hunting → agent window-server-2019 → Events; search data.win.system.channel: Security; Refresh	Q1	Events listed
5	DC	Get-Date	C4	Write the time down
6	DC	New-ADUser -Name "svc_backup" -SamAccountName svc_backup -AccountPassword (ConvertTo-SecureString "<Lab-Password>" -AsPlainText -Force) -Enabled $true	C5	No output
7	DC	Get-ADUser svc_backup	C6	Enabled : True
8	Wazuh	Search data.win.system.eventID: (4720 or 4722 or 4724 or 4726 or 4728 or 4732 or 4738 or 4756)	Q2	Rules 60109/60110
9	DC	Add-ADGroupMember -Identity "Domain Admins" -Members svc_backup	C7	No output
10	DC	Get-ADGroupMember "Domain Admins" | Select Name	C8	svc_backup listed
11	Wazuh	Search data.win.system.eventID: 4728	Q3	Rule 60159, level 12
12	Win 10	Sign in: Other user → HOPELABS\svc_backup. If refused, run C9–C17 (Part E)	—	Desktop loads
13	Wazuh	Agent DESKTOP-T6T1N2V; search data.win.system.eventID: 4624 and data.win.eventdata.targetUserName: svc_backup	Q4	Rule 60118 hits
14	Ubuntu	sudo cp /var/ossec/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.bak	C18	No output

#	Where	Command / action	Card	Expected
15	Ubuntu	sudo nano /var/ossec/etc/rules/local_rules.xml (paste Listing 1, Ctrl+O, Enter, Ctrl+X)	C19	File saved
16	Ubuntu	sudo /var/ossec/bin/wazuh-analysisd -t	C20	No ERROR lines
17	Ubuntu	sudo systemctl restart wazuh-manager	C21	No output; status active (running)
18	DC	Repeat steps 5–10 with svc_test2 (within 30 minutes)	C4–C8	svc_test2 in Domain Admins
19	Wazuh	Search rule.id: 100500 (agent window-server-2019)	Q6	Level 13 alert

Part E — If the workstation cannot sign in (domain unavailable)

1. `ipconfig /all` (C9): is the DNS server the DC on the host-only adapter?
2. `ping 192.168.56.115` (C10): replies mean the network is fine.
3. `nslookup HopeLabs.Local` (C11): "UnKnown 10.0.2.3" means the wrong DNS server answered.
4. `Get-NetAdapter` and `Get-NetIPAddress -AddressFamily IPv4 | Select InterfaceAlias, IPAddress` (C12, C13): find the 192.168.56.x adapter name.
5. `Set-DnsClientServerAddress -InterfaceAlias "Ethernet 2" -ServerAddresses 192.168.56.115` (C14) — replace the name with yours.
6. `Set-NetIPInterface -InterfaceAlias "Ethernet 2" -InterfaceMetric 5` and `Set-NetIPInterface -InterfaceAlias "Ethernet" -InterfaceMetric 100` (C16).
7. `ipconfig /flushdns` (C15), then `nslookup HopeLabs.Local` again. Expected: Server 192.168.56.115.
8. `systeminfo | findstr /B "Domain"` (C17): expect HopeLabs.Local. Try the sign-in again.

Part H — Cleanup and rollback

> **WARNING.** These cleanup commands are standard cmdlets and were **not executed** in the evidence. Read each one before running it and make sure the names are correct: deletion is permanent unless you restore a snapshot.

C22 Remove the test accounts from Domain Admins and delete them

Where / who	**Domain Controller, PowerShell** | Privilege: Domain administrator
Purpose	Undo the lab's privileged accounts.

Command	`Remove-ADGroupMember -Identity "Domain Admins" ` `  -Members svc_backup,svc_test2 -Confirm:$false`
	`Remove-ADUser -Identity svc_backup -Confirm:$false`
	`Remove-ADUser -Identity svc_test2 -Confirm:$false`

Breaking it down	**Part** / **What it means** `Remove-ADGroupMember` — Takes accounts out of a group; -Members lists them separated by a comma. `Remove-ADUser` — Deletes the account (this generates event 4726, which would also let you test deletion monitoring). `-Confirm:$false` — Skip the "Are you sure?" question.
Expected result	No output.
Observed in this lab	Not executed in the evidence.
How to verify	Get-ADGroupMember "Domain Admins" | Select Name shows only the intended accounts; Get-ADUser svc_backup reports not found.
If it fails	"Cannot find an object" → already removed or misspelt.

C23 Undo the Windows 10 DNS and metric changes (optional)

Where / who	**Windows 10, PowerShell** | Privilege: Administrator

C23 Undo the Windows 10 DNS and metric changes (optional)	
Purpose	Return the adapters to automatic settings if you want the original behaviour.

Command	`Set-DnsClientServerAddress -InterfaceAlias "Ethernet 2" -ResetServerAddresses`
	`Set-NetIPInterface -InterfaceAlias "Ethernet 2" -AutomaticMetric Enabled`
	`Set-NetIPInterface -InterfaceAlias "Ethernet" -AutomaticMetric Enabled`

Breaking it down	**Part** / **What it means** `-ResetServerAddresses` — Clears manually set DNS servers, so DHCP/automatic values are used. `-AutomaticMetric Enabled` — Lets Windows choose the metric again.
Expected result	No output.
Observed in this lab	Not executed. Note: undoing these brings the domain sign-in problem back.
How to verify	Run C9 and C11.
If it fails	Do this only if you do not need domain sign-in.

C24 Restore the original rules file	
Where / who	**Ubuntu VM, terminal** | Privilege: sudo
Purpose	Remove rule 100500 and reload.

Command	`sudo cp /var/ossec/etc/rules/local_rules.xml.bak \`
	`  /var/ossec/etc/rules/local_rules.xml`
	`sudo systemctl restart wazuh-manager`

Breaking it down	**Part** / **What it means** `cp backup original` — Copies the backup over the current file. `systemctl restart` — Reloads the rules.
Expected result	No output.
Observed in this lab	Not executed.
How to verify	Search rule.id: 100500 after a new test: no alert should appear.
If it fails	Skip this if you want to keep the rule.

Chapter 16 — Limitations and Future Improvements

- **Authored scenario.** The activity was generated by the lab owner, so "malicious intent" cannot be assessed.
- **Missing evidence:** Splunk output, Wazuh version, Agents page, the Ubuntu terminal screenshots, a 4720 record, a deletion test, and the user's sign-in screen after the fix.
- **Inconclusive item:** the DC search for 4624 on svc_backup (E-40).
- **Unexplained observations:** the ANONYMOUS LOGON 4738; the 13-minute alert delay on the workstation; the absence of a 60159 row for svc_test2 beside rule 100500; the changing server name printed by nslookup; an unrelated failed ping to 192.168.56.106.
- **Rule limits:** not account-specific, no allow-list, not tested for false positives or negatives (Section 7.3).
- **Photo quality:** several screenshots are camera photos; they were enhanced but remain smaller and less sharp than direct screenshots.
- **Effect on conclusions:** the detection claim (rule 100500 fires on the pattern) is solid for the one test run; broader claims about coverage and false-positive rate are **not** supported.

Chapter 17 — Conclusion

This lab investigated a rogue Domain Admin account end to end. The strongest verified findings are: the account svc_backup was created and enabled by Administrator on 29 October, promoted to Domain Admins at 11:15:48 (Wazuh rule 60159, level 12), and signed in interactively on DESKTOP-T6T1N2V on 31 October at 09:06:02. Post-logon alerts 61634 and 92052 were false positives, and an unexplained anonymous 4738 event remains open.

A custom rule (100500, level 13) now joins the creation and the promotion into one alert and fired on a second test account on 2 November at 12:14:04. A DNS-priority fault that blocked the domain sign-in was diagnosed and fixed.

The work shows practical skills in audit configuration, Active Directory administration, Wazuh triage, event interpretation, basic detection engineering and network troubleshooting. It does not prove Splunk correlation, deletion monitoring or false-positive performance. **Next steps:** add an allow-list and account-specific correlation, test deletion (4726), build the equivalent Splunk search, and clean up the test accounts.

Appendix A — Command History in Order

As used in the evidence. The lab password in the original commands is shown as a placeholder here.

```
# Domain Controller (PowerShell, administrator)
auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
auditpol /set /subcategory:"Security Group Management" /success:enable /failure:enable
auditpol /get /category:"Account Management"
Get-Date
New-ADUser -Name "svc_backup" -SamAccountName svc_backup -AccountPassword (ConvertTo-SecureString "<LAB-PASSWORD>" -AsPlain
Get-ADUser svc_backup
Get-Date
Add-ADGroupMember -Identity "Domain Admins" -Members svc_backup
Get-ADGroupMember "Domain Admins" | Select Name
```

```
# Windows 10 (PowerShell, administrator)
nslookup HopeLabs.Local
Get-NetAdapter
Get-NetIPAddress -AddressFamily IPv4 | Select InterfaceAlias, IPAddress
Set-DnsClientServerAddress -InterfaceAlias "Ethernet 2" -ServerAddresses 192.168.56.115
ipconfig /flushdns
nslookup hopelabs.local 192.168.56.115
Set-NetIPInterface -InterfaceAlias "Ethernet 2" -InterfaceMetric 5
Set-NetIPInterface -InterfaceAlias "Ethernet" -InterfaceMetric 100
ipconfig /flushdns
nslookup hopelabs.local
systeminfo | findstr /B "Domain"
```

```
# Ubuntu (Wazuh manager) - reported by the lab owner, no terminal capture
sudo cp /var/ossec/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.bak
sudo nano /var/ossec/etc/rules/local_rules.xml
sudo /var/ossec/bin/wazuh-analysisd -t
sudo systemctl restart wazuh-manager
sudo systemctl status wazuh-manager --no-pager | head -5
```

```
# Domain Controller - detection test (2 Nov 2026)
Get-Date
New-ADUser -Name "svc_test2" -SamAccountName svc_test2 -AccountPassword (ConvertTo-SecureString "<LAB-PASSWORD>" -AsPlainTe
get-aduser svc_test2
Add-ADGroupMember -Identity "Domain Admins" -Members svc_test2
Get-ADGroupMember "Domain Admins" | Select Name
```

Appendix B — Key Raw Values Quoted from the Evidence

Table 18. Key values

Item	Value
svc_backup SID	S-1-5-21-200352311-1354124744-2092920341-1129
svc_test2 SID	S-1-5-21-200352311-1354124744-2092920341-1130
Domain Admins SID (RID 512)	S-1-5-21-200352311-1354124744-2092920341-512
Administrator session on the DC	Logon ID 0x81e79
4728 event	record 67834, systemTime 2026-10-29T10:15:48Z, rule 60159 level 12
4738 event (anonymous)	record 67839, systemTime 2026-10-29T10:39:17Z, subject S-1-5-7, logon ID 0x3e6
4624 event	record 120739, systemTime 2026-10-31T08:06:02Z, logonType 2, targetLogonId 0x3c6be7e, targetLinkedLogonId 0x3c6be27
Sysmon dsregcmd event	UtcTime 2026-10-31 08:09:27.520, parent svchost.exe -k netsvcs -p
Rule 100500 alert	2026-11-02 12:14:04.697, level 13, rule.frequency 2, MITRE T1136.002/T1098

Appendix C — Glossary

Table 19. Glossary

Term	Plain-language meaning
Agent	Small program on a computer that sends its logs to the manager
Alert level	Wazuh severity from 0 (ignore) to 15 (critical)
Cmdlet	A PowerShell built-in command, named Verb-Noun (Get-Date)
Event ID	Number that identifies the kind of Windows event
False positive	An alert that looks bad but is harmless
Host-only network	A virtual network that only the VMs and the host share
Interface metric	A cost number; lower means the adapter is preferred
Logon ID	A number that identifies one sign-in session
Persistence	Attacker technique to keep access (for example a hidden account)
Privilege escalation	Gaining higher rights than before
Snapshot	A saved VM state you can return to

Appendix D — References

The links below were written from general knowledge without live web access; the pages may have moved or changed, so check each before relying on it.

- Wazuh documentation — https://documentation.wazuh.com
- Microsoft Learn — Windows security auditing events 4624, 4728, 4738 (search learn.microsoft.com)
- MITRE ATT&CK — T1136.002 Create Account: Domain Account; T1098 Account Manipulation; T1078 Valid Accounts; T1059.003 Windows Command Shell; T1055 Process Injection — https://attack.mitre.org
- Microsoft Learn — auditpol, Active Directory PowerShell module, NetTCPIP cmdlets